IMPROVED COOKSTOVE GROUPED PROJECT IN PAPUA NEW GUINEA

Project title	Improved Cookstove Grouped Project in Papua New Guinea
Project ID	4416
Crediting period	23-September-2024 to 22-September-2034
Original date of issue	14 June 2024 (date of registration)
Most recent date of issue	20 June 2025
Version	1.2
VCS Standard Version	4.7
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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Improved Cookstove Grouped Project in Papua New Guinea by TEM PNG Cookstoves Project Pte. Ltd. ("**TEM**") is a grouped project activity that involves the distribution of fuel efficient improved cookstoves (ICSs) within the boundary of Papua New Guinea (PNG). The ICS disseminated through this project will replace traditional cooking methods where open-fired wood-fueled cooking is the primary cooking method in the household.

The project activities included in this grouped project activity will occur in geographically defined regions within the project boundary of PNG, starting first within the Southern Highlands Province (SHP), Western Highlands Province (WHP), Jiwaka Province (JP) and Western, Eastern Highlands Province (EHP).

For more information on exact coordinates, refer to Section 1.13 of this document.

<u>A summary description of the technologies/measures to be implemented by the project:</u>

The technology of the project activity involves the use of ICS, whereby it is designed as a stainless-steel insulated combustion chamber, with a high thermal combustion efficiency (over 25%). This highly effective cookstove uses less biomass fuel and emits less smoke than open-fired wood-fueled cooking methods.

In the baseline scenario prior to the implementation of the project, there would be continued wood cutting practices and the use of non-renewable wood fuel by the target population to meet the thermal energy needs for cooking. With ICS technology, woody biomass would be more efficiently combusted, hence emitting lower carbon emission and in turn generating Greenhouse Gas (GHG) emission reductions. Additionally, the amount of wood cutting is estimated to be reduced, as well as the amount of time spent by end-users foraging for firewood.

Presently, a total of 19,206 ICSs were distributed, one to each household, within Phase I of the project, starting from 23/09/2024, and will eventually scale up to 100,000 ICS within the targeted Highlands Provinces in later phases. Given the success of these implementations, TEM may implement more projects and distribute more cookstoves. The ICS number will vary during actual implementation and same will update during registration and verification of project activity. The average annual and total GHG emission reduction (ER) from the project is expected to be **303,252 tCO2e** and a total of **3,032,518 tCO2e** for the group project over the entire crediting period.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	04/04/2023 – 14/06/2024	Verified Carbon Standard	Earthood Services Private Limited	-
Verification	23/09/2024 – 31/12/2024	Verified Carbon Standard	SustainCERT SA	

1.3 Sectoral Scope and Project Type

Sectoral scope[1]	Sectoral Scope 03 – Energy demand
Project activity type	Type II – Energy efficiency improvement projects

1.4 Project Eligibility

1.4.1 General eligibility

Justify that the project activity is included under the scope of the VCS Program and not excluded under Table 2.1 of the VCS Standard:

The project activity involves the introduction of energy efficiency and fuel-switch measures in cookstoves through the replacement of non-renewable biomass with more efficient project devices that use the same fuel as in the baseline, in turn generating GHG emission reductions and removals. Participation within this project is completely voluntary and participants may withdraw at any time, hence the project activity is in line with Section 2.3 of VCS Program v4.4, Program Scope.

The project boundary of the project activity is Papua New Guinea and does not involve electricity generation, infrastructure replacement (e.g. energy-efficient electric lighting or transformers) or installation of any kind, neither does it involve the recovery of waste heat to energy or to heat/cool via cogeneration or trigeneration. The project does not involve the production of thermal energy for industrial use and does not have any activities involved in switching from a higher to lower carbon content fossil fuel. The project activity also does not deal with the reduction of hydrofluorocarbon-23 (HFC-23) emissions.

Therefore, the project activity is confirmed to not be excluded under Table 2.1 of the VCS Standard v4.7.

[1] *Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes*

Pipeline Listing, Opening Meeting with Validation/Verification Team, and Validation Deadline:

The pipeline listing date of this project activity was 14/07/2023 and the start date of the project activity is 23/09/2024. This listing date is before the start date of the project activity; hence the pipeline listing is within the standardized method for determining additionality as per Section 3.8.7 of the VCS Standard v4.7.

History of this project activity:

The project activity was formally registered on 14/06/2024 under the methodology VMR0006: Energy Efficiency and Fuel Switch Measures in Thermal Applications, v1.2. On 09/10/2024, VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0 came into effect, replacing VMR0006 v1.2. It was stated that all VMR0006 projects must transition to VM0050 v1.0. starting with the 2027 vintage year at the latest[2].

Table 1 below details the project's listing, registration and methodology update process:

Table 1: Project Timeline Details

S/N	Date	Description
1	14/07/2023 – 13/08/2023	Project Listing Period
2	10/04/2023	Opening Meeting with Earthood Services Private Limited (1st Verifier)
3	09/11/2023	VCS Registration Requested with Verra
4	14/06/2024	Project Registered
5	31/01/2025	Opening Meeting with SustainCERT SA (2nd Verifier)
6	30/04/2025	Project Verification Requested

As per ***Table 1***, the project was listed on the Verra project pipeline before the opening meeting with the 1st Verifier, which represents the beginning of the validation process of the project activity. The validation process with the 1st Verifier concluded on 14/06/2024, after the 30-day public comment period. Therefore, the project timelines are compliant with the opening meetings criteria of Section 4.1.5 of the VCS Standard.

The project activity was also validated and registered before the start date of the project activity (23/09/2024), therefore it is compliant with the validation deadline set out in Section 3.8.1 of the VCS Standard.

Eligibility of Methodology & Double-Counting:

[2] VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0, https://verra.org/methodologies/vm0050-energy-efficiency-and-fuel-switch-measures-in-cookstoves-v1-0/

The methodology applied is VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0, developed under the Verra Standard. Under the applied methodology, double counting is avoided through the method of distribution wherein each ICS will carry a Unique Serial Number (USN) and the distribution of each device is logged into an electronic database which includes unique information for each recipient (i.e., Unique Serial Number, geographic coordinates, etc.) upon receipt of the device. The electronic database combined with internal checks done by the PP will ensure that no USNs are double counted.

1.4.2 AFOLU project eligibility

The project activity is not an AFOLU project therefore this section is not applicable.

1.4.3 Transfer project eligibility

The project activity is not seeking transfer from another standard; therefore, this section does not apply to the project activity.

1.5 Project Design

☐ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☒ Grouped project

1.5.1 Grouped project design

The grouped project activity involves energy efficient cookstove distribution which falls under the category of efficiency improvements in thermal applications; therefore it is eligible under the scope of VCS Program.

Eligibility Criteria

For the inclusion of new project activity instances, the Project Proponent (PP) shall ensure that each Project Instance (PI) is included:

Table 2: Evidence based on the Eligibility Criteria for Grouped project design

S/N	Criterion	Applicability	Evidence
1	Meets the applicability conditions set out in the methodology applied to the project.	New PIs will meet the applicability conditions set out in Section 4 of the methodology.	- Project Design Document.
f2	Use the technologies or measures specified in the project description.	Only ICS that conform with the grouped project description are to be distributed in the project.	- ICS manufacturer's technology description.

		The ICS will be chosen to deliver a level of service at least equivalent to the baseline appliance.	
3	Applies the technologies or measures in the same manner as specified in the project description.	ICS distributed in the PIs will adhere to the grouped project description.	- ICS manufacturer's Technical Specifications
4	Are subjected to the baseline scenario determined in the project description for the specified project activity and geographic area.	New PIs will be installed only in regions within the geographic borders of PNG subject to the same baseline scenario determined in Section 3.4 in this Project Description.	- Global Position System (GPS) coordinates captured from each end user will demonstrate the location of the project. - Virtual geographical log of cookstoves in monitoring database.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area. For example, the new project activity instances have financial, technical and/or other parameters (such as the size/scale of the instances) consistent with the initial instances, or face the same investment, technological and/or other barriers as the initial instances.	All new PIs will use the activity method for demonstration of additionality. **Step 1:** Regulatory Surplus There is no government mandated programme or policy in host country of this project ensuring the distribution of new project activity instances. **Step 2:** Positive List The inclusion of new project activity instances will comply with the positive list as it satisfies criterion 1 where it meets all the applicability conditions of the methodology.	- Analysis on PNG government programs or policies for cookstoves conducted in Section 3.5. - The End User Agreement (EUA) will confirm that the TEM "installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits".
6	Occur within one of the designated geographic areas specified in the project description.	New PIs will only occur within the boundary of PNG.	- GPS Coordinates of ICS end user.

7	Comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.	New PIs will comply with at least one complete set of eligibility criteria for the inclusion into the GPA.	- Project Design Document.
8	Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	New PIs will demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body through the inclusion of sufficient technical, financial, geographic, and other relevant information within the monitoring report.	- Sampling Plan - Monitoring Report
9	Be validated at the time of verification against the applicable set of eligibility criteria.	New PIs will be validated at the time of verification against the applicable set of eligibility criteria.	- Monitoring Report
10	Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	New PIs will provide evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance.	- The EUA
11	Have a start date that is the same as or later than the grouped project start date.	The start date of new PIs will be after 23/09/2024, after the start date of the GPA.	- ICS Distribution Ledger
12	Be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period (as set out in Section 3.4.4) and new instances are eligible for crediting from the start of the next verification period.	New PIs will be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). For example, when a PI starts with a fixed crediting period of 10 years (fixed), the project device/stove must be proven at inclusion that it will be able to sustain	- Monitoring Report - Project Device Lifespan Specification

		through the crediting period. This may be proven through the Project Device Lifespan Specification document.	
13	Only eligible for crediting from the start of the verification period in which they were added to the grouped project.	Crediting Periods of New PIs will only start after the date that the project has been included in the grouped project.	- Monitoring Report
14	Not be or have been enrolled in another VCS project.	New PIs must declare that it has not or have been enrolled in another VCS project.	- Project Letter of Undertaking
15	Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 - 3.6.9 of the VCS Standard.	New PIs will adhere to the clustering and capacity limit requirements set out in Section 3.6.8 – 3.6.9 whereby the Project participant (PP) shall include all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent and where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit.	- GPS Coordinates of ICS end user.

1.6 Project Proponent

Organization name	TEM PNG Cookstoves Project Pte. Ltd.
Contact person	Dr. Gabriel Machovsky Capuska
Title	Director of International Projects
Address	15th Floor, 31 Market Street, Sydney, NSW 2000, Australia
Telephone	+61 3 5977 1286
Email	gabriel@tem.com.au

1.7 Other Entities Involved in the Project

TEM PNG Cookstoves Project Pte. Ltd. is the sole entity involved in the project and no other entities were involved in the project activity, therefore this section is not applicable.

Organization name	
Role in the project	
Contact person	
Title	
Address	
Telephone	
Email	*The email address domain must match that of the organization.*

1.8 Ownership

The project ownership is with TEM PNG Cookstoves Project Pte. Ltd.

TEM has also consulted with national authorities such as the PNG Forest Authority (PNGFA), wherein a Letter of Non-Objection to the project activity has been received.

During the registration of ICS, the participating households will sign an agreement to transfer the ownership rights of the carbon assets generated from this project to the project proponent.

1.9 Project Start Date

Project start date	23-September-2024
Justification	First End-User Agreement signed with end-user. The project activity was also registered on 14/06/2024, well within the VCS Standard's requirement of two years of the project start date.

1.10 Project Crediting Period

Crediting period	☐ *Seven years, twice renewable* ☒ *Ten years, fixed* ☐ *Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)*

Start and end date of first or fixed crediting period	23-September-2024 to 22-September-2034

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☐ < 300,000 tCO2e/year (project)

☒ ≥ 300,000 tCO2e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
2024	103,526
2025	372,694
2026	354,061
2027	336,357
2028	319,539
2029	303,562
2030	288,385
2031	273,965
2032	260,268
2033	247,254
2034	172,906
Total estimated ERRs during the first or fixed crediting period	**3,032,517**
Total number of years	**10**
Average annual ERRs	**303,251**[3]

[3] *Note: A difference of +/- 1 in ERs is due to the rounding from project years to vintage years*

1.12 Description of the Project Activity

The project involves distribution of fuel-efficient ICS to replace the baseline cookstoves in households. Prior to the implementation of the project activity, households use non-renewable wood biomass for their fuel needs. The baseline appliance being replaced by the project is an open fire (typically based on the three-stone concept).

List of facilities, systems, and equipment in operation under the existing scenario prior to the implementation of the project:

In the baseline scenario, no improvement projects relating to efficiently combusting woody biomass for cookstoves had been implemented, as such, wood fuel use has steadily been increasing as economic pressure has been increasing. The distance to wood harvesting sites increases ever more as fuel demand increases.

The ICSs reduce GHG emissions from biomass burning through improved combustion efficiency of wood fuel and decreased wood fuel consumption. This is achievable through improved heat transfer to cooking pots through the insulated walls and design of the cookstove top as compared to the baseline scenario and traditional cooking methods over open- and three-stone fires, wherein woody biomass would be burnt at a lower efficiency, incurring higher GHG emissions.

List and arrangement of main manufacturing technologies and equipment involved:

The project envisages to use the highest performance ICS in the market known as G3 Burn rocket cookstove with an adjustable pot skirt (see Figure 1 and Table 3).

G3 Burn Rocket Stove
with adjustable pot skirt

Figure 1: G3 Burn rocket cookstove (also known as Kuniokoa or Ecoa wood cookstove) with adjustable pot skirt.

Table 3: Technical description of G3 Burn rocket cookstove based on manufactured specifications.

Parameter	Description

Thermal efficiency with pot skirt	52.0%[4]
Efficiency	Tier 4
Emissions	Tier 3-4
Dimensions	32.1 cm height, 28.2 cm diameter
Construction materials	▪ Stove Body: Galvalume ▪ Pot Rest: Stainless Steel ▪ Pot Skirt: Stainless Steel ▪ Burning Chamber: Stainless Steel ▪ Fuel Feeding Door: Stainless Steel ▪ Stick Shelf: CRCA ▪ Legs: Aluzinc
Fuel	Firewood, bamboo, crop residue, other type of biomass
Life Span	10 years

<u>The types of and levels of service provided by ICSs:</u>

Level of Service:
Project Instances will provide ICS based on the G3 Burn rocket cookstove specifications as stated above, delivering a level of service at least equivalent to the baseline appliance (open fire used for household cooking). The ICS distributed under Project Instance (G3 Burn Rocket Cookstove, also known as Kuniokoa or Ecoa wood cookstove) will be specified in the monitoring report.

Type of Service:
The G3 Burn rocket stoves will deliver a specified high-power thermal efficiency of at least 25% or greater and evidenced by a manufacturer's specification and WBT certificate. This is a significant improvement on the baseline open fire thermal efficiency default of 10% (as per AMS II.G).

Data collection of ICS end-user

Project proponent must gather the necessary information to identify households using its ICS during the project. To facilitate this process, each ICS will be assigned a USN. This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address / GPS of ICS household
- Phone number of ICS user or household, where available.
- Stove model
- Date of distribution/installation
- ICS serial number
- Retailer/distributer information

The information collected will be stored in which will serve as project database for project monitoring and sampling purposes.

1.13 Project Location

[4] Reference document to validator: WBT test report by CREEC Laboratory

The GPA will distribute ICS to households within the project boundary of PNG (Latitude: 6.3150° S Longitude: 143.9555° E). The first project instance is projected to be within the Highlands Region (Figure 2 , and will include the following areas: SHP, WHP, EHP and JP (see Figure 2 & *Table 4)*. Based on the PNG census 2011, the total population size of these four Highland Provinces was 1,792,931[5] and it is estimate that has substantially increased.

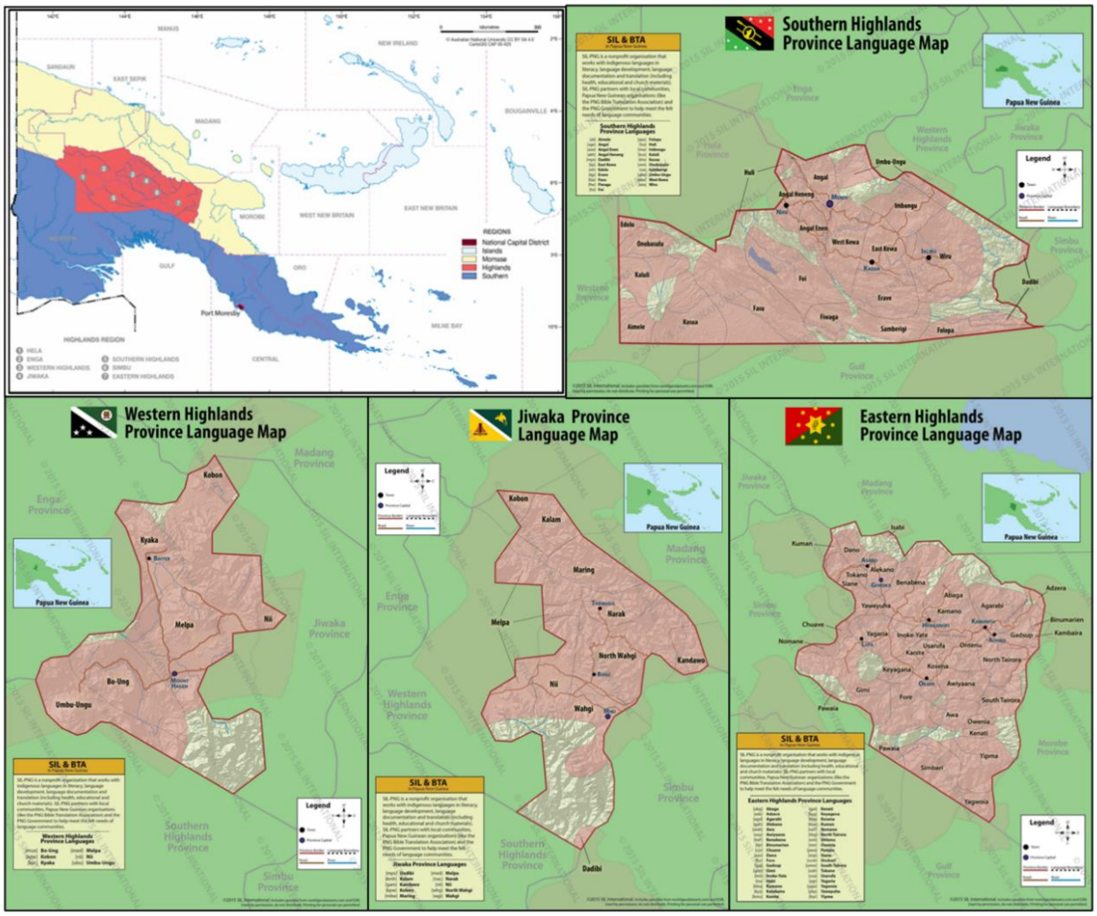


Figure 2: Map of Papua New Guinea including the Highlands Region (top left) and maps of the four different provinces included in the first project instance, Southern Highlands Province (top right), Western Highlands Province (bottom left), Jiwaka Province (bottom center) and Eastern Highlands Province (bottom right). Maps extracted from www.pnglanguages.sil.org

[5] National Statistics Office, Papua New Guinea National Population & Housing Census 2011, 'Count me in' (2011)

Table 4: The Geographic coordinates of the four provinces involved as the first project instances of the group activity in Papua New Guinea.

S/N	Location	Latitude	Longitude
1.	Southern Highlands Province	6.4179° S	143.5636° E
2.	Eastern Highlands Province	6.5862° S	145.6690° E
3.	Western Highlands Province	5.8310° S	144.1720° E
4.	Jiwaka Province	5.8564° S	144.5218° E

1.14 Conditions Prior to Project Initiation

The conditions prior to project initiation are the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

The pre-project devices are inefficient three stone fired or conventional systems lacking improved combustion air supply mechanisms and flue gas ventilation system i.e., traditional stoves. The baseline scenario is the same as the conditions existing prior to the project initiation i.e., continue the use of traditional cook stoves (Please refer to section 3.4 – Baseline Scenario).

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of ICS in PNG. The project is a voluntary effort by the project proponent.

There is no negative concern made on the ICS project from the above law.

Additionally, TEM maintains a continuous communication with PNG's National Authorities providing frequent updates on the project activities. As an example, on the 31/01/2023, TEM had consulted with PNGFA and Climate Change and Development Authority (CCDA). During the consultations, the draft PDD and Local Stakeholder Consultation Plan were provided to the leadership of the government bodies for their comments. However, no comments have been received.

The acknowledgment letters from the abovementioned authorities can be made available to validators upon request.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 0 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

<u>**Response:**</u>

The project activity involves the distribution of ICSs to households within PNG. The boundary of the emission reductions is limited to the decrease in firewood consumption of each household, hence there are no indirect upstream or downstream GHG emissions to the organization's supply chain.

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The project will enable and enhance households to achieve several Sustainable Development Indicators (SDI):

The project provides ICSs with high thermal efficiency to reduce the consumption of firewood which achieves an efficient use of non-renewable natural biomass. This impacts the estimated reduction in removal of woody biomass to the tune of ~3.70 tons of firewood per cookstove per year, from forests surrounding the communities thereby leading to an increase in above-ground biomass in these forests. (SDI15.2.1)

The project activity will replace 100,000 traditional cookstoves with fuel efficient ICSs. The benefits of changing from a traditional cookstove to the more efficient ICS results in the consumption of lesser firewood, that is usually purchased at high prices, and cleaner cooking. Hence, this reduces the economic burden of families under the project activity (SDI Goal 1) and ensures households have access to clean cooking (SDI Goal 7).

The ICS reduces the levels of fine particulate matter ($PM_{2.5}$) emissions in the air compared to the baseline emission level of 943.3 mg/MJ, thus, improving the health and well-being of the end-users involved in the project. (SDI Goal 3)

With the project activity, jobs are generated for house registration and ICS distribution for participants. Also, additional jobs for the local community during sampling for verification, for the maintenance of cookstoves and various project activity will be required throughout the project lifetime. This creates long term jobs for the local community, bringing opportunities and inclusiveness to people from different genders, age, occupation, and persons with disabilities. TEM envisions that 50% of the project's employees will be female. (SDI Goal 8 and SDI8.5.1)

With the creation of jobs, the project proponent will also develop training activities for vocational and relevant skills of local individuals by providing non-formal education and training on issues related to climate change, with specific skill building in operations and surveying activities related to the ICS distribution and its monitoring under VCS. (SDI Goal 4.1 and SDI Goal 4.2)

Lastly, there is a contribution to the GHG emission reduction through an estimated reduction of ~3.2 tCO_2e per ICS due to the replacement of baseline cookstoves with Burn Cookstoves over the crediting period. (SDG 13)

The UN Sustainable Development goals were adopted by all member countries of the United Nations in September 2015. These goals are closely aligned with the 'PNG Vision 2050' and the National Strategy for Sustainable Development known as 'StaRS which have now become the focus of sustainable development activities in the country[6]. The PNG Vision 2050 outlines the need for economic growth in local communities through skill development, reduction of greenhouse emissions and conservation of biodiversity. Regarding StaRS, this project aligns with the strategy advocated for sustainable economic growth while preserving the environment around.

Finally, this project also aligns perfectly with the PNG National Health Plan for 2021-2030. The PNG Government's vision is for a healthy and prosperous nation where health and wellbeing are enjoyed by all. The goal is to prevent ill health, identify and address health risks and emerging diseases, and provide accessible and affordable quality healthcare for all. This is aligned with the proposed project activities causing the reduction of fine particulate matter ($PM_{2.5}$) emissions within households which addresses the health risks related to the fine particulate matter.

[6] PNGDNPM. (2021). *Papua New Guinea Vision 2050 | Papua New Guinea Environment Data Portal*. https://png-data.sprep.org/dataset/papua-new-guinea-vision-2050

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

According to the applied methodology, leakage emissions (LE_y) associated with reduced or avoided use of non-renewable biomass is accounted for using an adjustment factor of 0.95 is applied to the GHG emission reductions in Equation (11) (i.e., $0.95 \times (BE_y - PE_y)$).

As the project activity does not involve the use of renewable biomass ($LE_{RB,y}$), leakage emissions calculated using CDM Tool 16 v5.0 as per the applied methodology will not be used and will remain as '0' within the emission reductions calculations.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

Not applicable.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	Under this project, the stakeholders are identified as <u>local people</u>, <u>communities</u> and/or representatives who are directly or indirectly affected by the activities proposed. Examples of identified stakeholders are the <u>end-users</u> and <u>cookstove manufacturer.</u> **Aligned with the United Nations Convention on the Rights of Indigenous People (2007)[7] and the United Nations Development Program (UNDP) Stakeholder Engagement Guidelines (2022)[8], the development of a project encompasses engagement with multiple stakeholders at different stages that spans from Project Identification to Implementation.**

[7] https://www.ohchr.org/sites/default/files/Documents/Issues/IPeoples/UNDRIPManualForNHRIs.pdf

[8] https://ses-toolkit.info.undp.org/sites/g/files/zskgke446/files/SES%20Document%20Library/Uploaded%20October%202016/UNDP%20SES%20Stakeholder%20Engagement%20GN_Final_rev_July2022.pdf

Legal or customary tenure/access rights	The project aligns with several International and Papua New Guinean's laws, regulations and conventions. TEM, as the project developer, and its in-country team will follow and obey all these relevant laws, regulations and conventions (as stated below): International Laws, Regulations and Conventions: • United Nations Framework Convention on Climate Change (UNFCCC,1992) • United Nations Declaration on the Rights of Indigenous Peoples Rights (2007) • Stockholm Declaration of the United Nations Conference in Human Environment (1972) • Rio Declaration on Environment and Development (1992) • Paris Agreement (2015) Papua New Guinea's Laws, Regulations and Conventions: • Papua New Guinea's Constitution (1975) • Climate Change Management Act 2015, Amended (2022), and proposed Carbon Market draft regulation (2023) • Forestry Act (1991) • Environmental Act (2000) • Investment Promotion Act (1992) • Industrial Relations Act (1962) • Industrial Health Safety and Welfare Act (1962) • Employment Act No. 54 of 1978 (Consolidated up to 31 March 2001) • Child Welfare Act (1961) • Discriminatory Practices Act (1963) • Workers Compensation Act (1978) • Papua New Guinea Vision (2050)
Stakeholder diversity and changes over time	Social Capital: Papua New Guinea (PNG) is characterised by a high degree of income inequality with 40% of people living below the extreme poverty line and 41% of children living in poverty[9]. Lack of education, healthcare and infrastructure are also known as the main

[9] https://data.unicef.org/how-many/how-many-children-live-in-poverty/

	contributors to influence poverty. In addition, 80% [10] of the population lives in rural areas without the benefits of electrification, vaccinations, or ICSs. It is estimated that the country's rural population increased 1.8% from 2021 to the 8,765,048 measured in 2022[11]. <u>Health and Well Being:</u> In accordance with the World Health Organization's (WHO) guidelines, the air quality in PNG is considered moderately unsafe, with the annual mean concentration of PM2.5 exceeding the recommended maximum of 10 µg/m3[12],[13]. Vehicle emissions, the mining industry, the oil and gas industries, and waste burning are suggested as the main drivers for this pattern. This is also magnified by indoor air pollution and smoke generated from the burning of solid fuels by traditional open fire cooking and heating that is linked to various illnesses and deaths[14]. The Baseline Study (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf) showed that open fire cooking (i.e. inefficient three-stone fire burning non-renewable woody biomass) produced thick grey or light blue smoke that led to headaches, cough, asthma, and other related diseases. <u>Economic Capital:</u> The PNG population is largely rural and growing with around 60% of the population being of working age (15-64 years) as shown in Figure 3 below:

[10]https://www.mypsup.org/countries/Papua_New_Guinea#:~:text=Some%2080%20per%20cent%20of,country%20has%20several%20urban%20centres.

[11] https://www.macrotrends.net/global-metrics/countries/PNG/papua-new-guinea/rural-population#:~:text=Papua%20New%20Guinea%20rural%20population%20for%202022%20was%208%2C765%2C048%2C%20a,a%202.06%25%20increase%20from%202019.

[12] World Health Organization (WHO), Health and environment scorecard: Papua New Guinea, Air Pollution

[13] https://www.who.int/en/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health

[14] World Health Organization (WHO), Indoor smoke from solid fuels

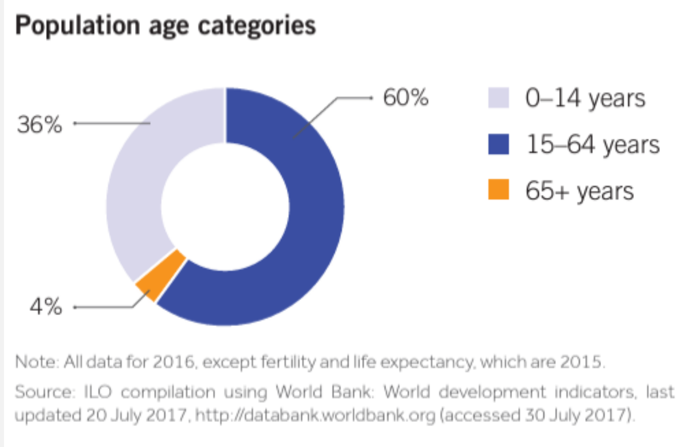


Figure 3: Population age categories, depicting age groups that are too young to work (0-14 years), within a typical working age (15-64 years), and retired (65+ years).

According to Employment and Environmental Sustainability Fact Sheet (2017), the PNG labour force participation rate was 70.5%, and the employment-to-population ratio was 68.7%, both close to gender parity. In addition, the vulnerable employment in the country accounted for 44.2% of the labour force, with many of those workers having an own-account status. The split between Vulnerable employment, Employees, and Employers is shown in Figure 4 below:

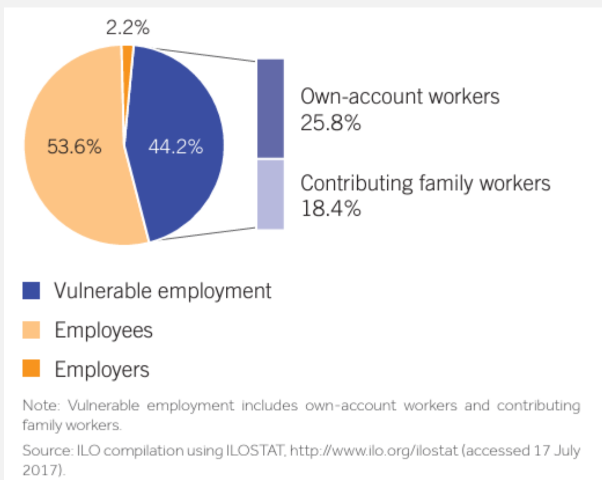


Figure 4: Presentation of Vulnerable Employment in Papua New Guinea by status in 2017

Own-account and contributing family workers are more likely to experience low job and income security, and lower coverage by social protection and employment regulations than employees and employers.

Informal sector employment is particularly widespread among the youth. A gender gap in access to formal sector employment means that men are more likely to find formal jobs than women. Wages in

	the informal economy are considerably lower than in the formal economy[15]. This pattern is particularly relevant to the rural areas of the Highlands region that is characterized by low and informal income, mostly derived from the sales of agricultural produce (e.g., fresh food, coffee and firewood) in street markets (Figure 4), with 40% of the population living near the poverty line (Hanson, et al., 2001[16]) and an average income estimated as K439/month (Bonilla, et al., 2020[17]).
Expected changes in well-being	The distribution of ICSs within the project boundary is expected to reduce the amount of firewood used for cooking and heating by the end-users. It is also expected that the local communities will incur less trips per week to collect firewood. In those households where firewood is purchased, an increase in fuel efficiency will result in monetary savings during the project lifetime. The continued usage of the ICS by local communities will also gradually reduce their exposure to high CO and PM2.5 and due to higher combustion efficiency of this new technology over the lifetime of the project activity. The project will aim to hire around 130 local people through a fair and unbiased process, envisioning 50% gender equality over the project lifespan.
Location of stakeholders	The stakeholders are located within the project boundary of Papua New Guinea, where the project activities will distribute ICS to communities within the Highland region: Southern Highlands Province (SHP), Western Highlands Province (WHP), Eastern Highlands Region (WHP), and Jiwaka Province (JP).
Location of resources	The project activity involves the distribution of ICS to individual households and does not consider any ownership over an end-user's territory, therefore this section is not applicable to the project activity.

[15] Improved Technical and Vocational Education and Training for Employment Project: Report and Recommendation of the President, https://www.adb.org/projects/documents/png-53083-001-rrp

[16] Hanson, L., Allen, B., Bourke, R. (2001). Papua New Guinea Rural Development Handbook. Canberra, Australian National University

[17] Bonilla, F., Voorend, K., Anker, R., Anker, M., & Prates, I. (2020). Anker Living Income Reference Value: Rural Papua New Guinea (2020) (No.20-03-20). Universidad Privada Boliviana.

2.1.2 Stakeholder Consultation and Ongoing Communication

Baseline Study Workshop and Trainings

Prior to conducting the baseline survey, a total of 26 local field assistants and coordinators were trained as seen in Figure 5 on: i) data collection techniques, using accurate survey techniques that are based on integrity and ethical standards, ii) measure, and identification of firewood species and food sources used for daily consumption and iii) quantify the level of smoke in open fire cooking practices using observational techniques . These field assistants and coordinators consisted of primary school teachers, including students and nursing officers who lived among the communities where the households were surveyed in lingua franca Tok Pisin.

Figure 5: Training of local field assistants and coordinators by Project Proponent

During the training, the field assistants acquire information on the carbon cycle, climate change challenges and mitigation strategies, and the use of carbon finance on developing sustainable benefits to local communities through carbon projects.

Engagement with Broader Communities

Following the training workshops, the baseline survey was conducted in November 2022, spanning a period of 3 weeks. The survey incorporated two techniques for data collection: survey questionnaires and control cooking tests (CCTs). The former was used to collect general details such as personal information, cookstove and cooking information, baseline firewood information and food source types for consumption. The latter involved measurements of firewood and foods using both the baseline (open fire cooking practices) and ICS shown in Figure 6. During this process, firewood and food measurements were taken before and after cooking using scale weights to ensure an accurate measure of both firewood and food consumption.

Figure 6: Preparation of food (left), Controlled Cooking Test (right)

The duration of cooking and level of smoke produced (e.g., thick, thin, light grey and medium thick,) while cooking was also recorded to understand whether the ICS provided by the project lead to the reduction of time spent cooking and toxic smoke produced within the dwelling.

For more information, please refer to the file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf, and the audited file titled: SPFEC Letter to TEM-Review of Cookstoves Highlands region report_17-01-2025.pdf).

Date of stakeholder consultation	Southern Highlands Province: 13 March 2023 Western Highlands Province: 15 March 2023 Jiwaka Province: 17 March 2023 Eastern Highlands Province: 20 March 2023
Stakeholder engagement process	Prior to the LSCs, TEM utilized several media outlets to advertise the event to the provinces. These efforts were fulfilled using: • Newspaper Advertisements • Letters of Notification to Provincial Governments • Radio Broadcasts • Invitation Letters to Churches • Social Media Posting • Webinar • Email to PNG Australia Alumni Association (PNGAAA) members All advertising outlets were run in a staggered format aiming to keep stakeholders informed and aware of the LSCs meetings happening within their provinces. For the exact dates of advertisements being published, please refer to Appendix II: LSCs Outreach. The stakeholders engaged in this project were people or groups of people that were directly or indirectly affected by the project as well

as those who were interested in the project and or had the ability to influence its outcome either positively or negatively. They include:

- Representatives from the provincial administration
- Representatives from Women Groups
- Representatives from Churches
- Ward councillors
- Heads of households both male and female
- Key informants such as teachers, community health workers etc.
- Members of communities within project boundary that meets the project's household selection process
- Youth leaders and
- Educated elites (lecturers, nursing officers, Health Extension Officers, Police personals etc.)

Southern Highlands Province LSC

A total of 159 people attended this LSC on the 13th of March 2023 held at Kumin Catholic Mission in Mendi, SHP. This event included the presence of public servants such as nursing officers, police, teachers and pastors that commuted into Mendi from various neighbouring districts.,

Figure 7: A demonstration by Ms. Magret Yangpin and Mr. Richard Manda – SHP Field Assistants.

The LSC was facilitated by Mr. Michael Kisombo, the project Manager with Simi Oa Consultancy Services who was engaged by TEM, through a power point presentation after welcoming and acknowledging all the stakeholders that were present. The presentation was prepared in English, and translated in *tok pidgin*, as the common local language in the region.

Attendees were briefed on the purpose of the meeting, and informed of the:

- Carbon cycle and the important role it played within their forests.
- Findings of the Baseline Study conducted in the Highlands region including SHP and WHP (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf)
- The Verra registration process
- The role of carbon finance in the project activity
- Indicative timeline and implementation plan of the project

During the LSC, two field assistants as seen in Figure 7 were then invited to demonstrate and to explain how the ICS is used. An additional end-user from the crowd was also invited to give her experience on the benefits her family is experiencing while using the ICS.

After the presentation, stakeholders were given the opportunity to ask questions and provide their reflections. For the summary of questions asked and subsequent answers provided by TEM, please refer to Appendix III: Stakeholder Feedback.

After the discussions, there were about 29 feedback forms that were received through the feedback forms that may be made available to validators upon request.

Western Highlands Province LSC

A total of 61 people attended this LSC on the 15th of March 2023 held at Kuri Lodge in Mt Hagen, WHP. Participants who attended came from different backgrounds and districts. The LSC was also covered by Radio Eagle (local radio station) that was present all throughout the presentation.

Figure 8: Group photograph at the Western Highlands Province LSC

Prior to the introductions and presentation, Pastor Enoc Wan of Hagen Central – Pentacoastal church opened the session with a word of prayer. Mr. Michael Kisombo then introduced the project and invited his field assistants to introduce themselves. He then proceeded with the presentation using *tok pidgin*.

During the presentation project activity was introduced including an overview of climate change challenges, the carbon emissions and the role of forests in restoring carbon and the impacts if most forests are destroyed. Following the background of the project, he explained the baseline study conducted in WHP and the results attained from the survey and invited end users (from the baseline study) present to explain how they are currently using the ICS to date and their impact on them.

To note, the presentation was not only translated to tok pidgin but also made more accessible making concepts more amenable to the broader audience. The change was made upon reflection of feedback received during the SHP LSC where participants asked for technical concepts to be broken down to commonly used languages and analogies. For example, the concept of carbon emission destroying the ozone layer; Mr. Kisombo would explain it as ‘bad air that destroys the blanket protecting sun and earth “and so on.

Two volunteers in the group gave their reflections and how they are using the cookstove. Both spoke highly of the impacts the ICS had on their families. Michael then asked Tony (one of the fields assistants) to demonstrate how the ICS is used. After that presentation, Mr. Kisombo proceeded with the explanation of the Verra registration process and role of carbon finance and continued to explain the large-scale project, its indicative timeline and implementation schedule.

After the presentation, stakeholders were given the opportunity to ask questions and to give their reflections followed by a closing prayer by Pastor Enoc. 29 feedback forms were received through the feedback forms out of the 61 participants.

Jiwaka Province LSC

More than 90 people attended this LSC on the 17th of March 2023 held at Jiwaka Mission Resort (JMR) in Banz, Jiwaka. During initial consultations to invite the provincial administrator to the LSC, the provincial administrator showed great support for the project and was able to communicate widely to his people to participate. Due to work commitments, he sent his Deputy Administrator to attend the local stakeholder consultation instead.

Figure 9: Presenter Michael Kisombo (right) and the Deputy Administrator (Left)

Mr. Michael Kisombo led the discussions with a welcome and thanked everyone who came to Jiwaka Mission Resort for the local stakeholders meeting. He also did the presentation in *tok pidgin* – aided by the slides on power-point that was used in two other provinces. Similar to past LSCs, the topic on carbon cycle and the destruction of the ozone layer was simplified to the level of the participants. This topic was further elaborated on and supported by a few participants who had wider experience working or teaching in schools in this area of specialty.

The baseline in SHP and WHP and the coordinators of these projects, Mr. Patrick Wapiyu and Arnold Terry, were also on hand to explain the results of the baseline study and demonstrate how the ICS was used. After these discussions, Mr. Kisombo continued with the presentations on the purpose of the stakeholder meeting, the verra registration process, carbon finance and the sustainable goals that this project is meeting. Towards the end, the indicative timeline and implementation schedule was explained before allowing for question-and-answer session.

There were about 29 feedback forms received through the feedback forms out of the 90 registered participants.

Eastern Highlands Province LSC

Around 29 people attended this LSC on the 20th of March 2023 held at Steakhouse Goroka in Goroka, EHP as shown in Figure 10. The low turn-out rate may be attributed to the project being very new to the local stakeholders. Most stakeholders who attended learnt of the event word of mouth or through friends' network and face-book post. Despite a lesser number the discussions were fruitful. Almost all agreed that they see there is no harm and supported the project.

	 Figure 10: Community members involved in the EHP LSC at Steak House Goroka. Similar to the first three LSCs, the presentation was conducted by Mr. Michael Kisombo in *tok pidgin* – as the common language where most could understand well. Michael followed the same presentation he did in the other three provinces. The sample ICS that was used for this LSC was displayed and a demonstration was done on how the ICS would be efficiently used. Although it was the first presentation made to the stakeholders in the Eastern Highlands Province, everyone who attended was positive with the project and supported the project activity. After the discussions, there were about 24 feedback forms received from the 29 participants. At the end of the LSCs, stakeholders were informed of the project's email (pngcookstoves@tem.com.au), as well as direct contact details to the local project manager and lead project manager in TEM. In all marketing documents, the project email and website were also published, to ensure stakeholders had multiple avenues to reach out to TEM with feedback, allowing for on-going communication.
Consultation outcome	There was no negative feedback received during the LSC. Most questions aimed to seek clarity on how a more efficient cookstove could reduce carbon in the atmosphere, when the project could begin, and if more cookstoves could be distributed. For a complete list of questions and feedback, please refer to Appendix III: Stakeholder Feedback. For more information, a Local Stakeholder Consultation Report is available upon request of the validator.
Ongoing communication	During the LSC, it was discussed how stakeholders may keep in contact with TEM's in country project manager, Michael Kisombo. During which, the following solutions were offered: 1. Reaching out to the in-country Team during the twice a year visits to maintain the ICSs.

	2. Reaching out to TEM through the project's dedicated email. 3. Reaching out to TEM through the project's website. The following website of the project activity was available to the stakeholders: https://www.tasmanenvironmental.com.au/pngcookstoves/ The following email addresses will be also used for inquiry or comment on the project: pngcookstoves@tem.com.au TEM in-country Project Manager: Michael Kisombo, michael.kisombo@tem.com.au. TEM's Office in Papua New Guinea (TEM PNG) Number: +675 71532292
Stakeholder input	Please refer to Appendix III: Stakeholder Feedback.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The project is voluntarily implemented by TEM PNG Cookstoves Project Pte. Ltd., and end-users are free to choose whether they will be involved in the project or not. Free, prior, and informed consent takes place during household registration and cookstove distribution through signing off that they would like to be part of the project, have received the ICS, and they will transfer the rights of the carbon credits generated from the project to TEM PNG Cookstoves Project Pte. Ltd. End-users are free to withdraw from the project activity at any point in the project lifetime. **Steps of obtaining consent:** • TEM's has multiple operational field teams and each of them consists of 4 members (1 team leader, 2 field assistants and 1 cookstove ambassador). These teams visit each household following Free, Prior Inform Consent (FPIC protocols) to seek consent from end users to participate in the project. • Once consent is obtained, the details of the end user are collected digitally by the field assistants via a tablet and recorded in a Household Registration Form and a token flyer is provided that has information on project benefits and a unique token number. • The project benefits and the contents of the End user Agreement are translated and explained by the field team during the house visit. • The End user Agreement was translated and explained a second time by TEM's field team during the cookstove handover to the end user before signing. **Token Number and Unique Serial Number**

	The unique token number provided to the households during the time of registration and the unique serial number (USN) are different identifiers used to ensure that only one improved cookstove is distributed to a household. A Token flyer with a unique token number is provided to the end user after consent is obtained to participate in the process and the household registration step is completed. This token flyer exchanged for an improved cookstove during the distribution of the improved cookstove after the end user is verified with the photo taken during registration. The USN of the improved cookstove is logged against the token number in the data management database, which ensures that only one device is allotted per household.
Outcome of FPIC	For end-users to be considered part of the project, their consent will be sought at two major points of the project onboarding process: 1. Household Registration 2. ICS Distribution During the household registration phase, the end-user's consent to participate in the project is obtained. Once the willingness to participate in the project is acknowledged, information on the house geographical location and alongside a list of prerequisite questions relating to their baseline cooking behaviors are collected. The end-user is then made aware of what the project entails, how to collect their ICS, and who they may need to contact if they have any questions before they sign off on a Household Registration Form, marking their consent to join the project activity. They are then given a Token Flyer that contains an individual code that is linked to their house and name, which they will bring down to the ICS distribution site to collect their cookstove. Field assistants will take a picture of the end-user with their token and individual code to be stored in our database. During the ICS distribution phase, the end-user presents the token to a field assistant who checks the token code against TEM's database to match the end-user's face to a picture within the computer system. Before the ICS is dispensed at no cost to the end-user, the field assistant briefs the end-user on how to use and maintain the ICS and the environmental, health and economic benefits of using the ICS. Once that is acknowledged, the field assistants explain the EUA stating that the carbon offsets generated by that individual improved cookstove will be transferred to TEM. Upon the end-user's consent, the EUA is signed, and the end-user is given the cookstove at no cost. As the project activity involves the distribution of ICS to individual households, no land has been encroached or has there been any relocation of people without consent nor forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

Development process	The Grievance Redress Mechanism (GRM) is a community driven approach built on traditional Melanesian customary protocols, that are formalised to accept, assess, and resolve feedback or complaints regarding the implementation of the project. The different channels to report grievances are summarised in **Table 5**, whereas the process of organising common types of grievances is captured in **Table 6.** The Procedure recognises and prioritises the traditional and customary means of conflict resolution in Melanesian society by encouraging dialogue through clan leaders and staff at the community level, and utilising Field Assistants who are sourced from local communities to act as the first responders while addressing grievances. Outcomes of the GRM are recorded in the GRM logbook. These measures are taken to enhance Project accountability and transparency and to support the project initiatives.
Grievance redress procedure	Table 5: Channel for Grievance Reporting (see below); Table 6: Process of organising common grievances (see below)

Table 5: Channel for Grievance Reporting

Grievances Reporting Channel	**Grievance Reporting Mechanism**
Physical walking in	Complaints to TEM PNG Office and/or its staff
Social media and online	Complaints made online to TEM PNG Office, and/or in country Project manager
Phone call	Phone calls to Field Assistance, and/or Project manager
Verbal through community representatives	Complaints through community representatives who then channel through the organizational structure

Table 6: Process of organising common grievances

Type of Grievance	**Scope**	**How will be facilitated**
Unfair distribution of ICS	Cookstoves unfairly distributed by project staff	Gather information, investigate and refer to project lead
Burglary, sold or destroyed by natural disaster	Cookstoves were stolen, destroyed, or sold	Gather information, investigate, and refer to project lead
Household missing out on the distribution	Cookstoves missed targeted households	Investigate and refer to project lead for possible re-issue
Damaged cookstoves	Cookstoves issued were damaged or	Investigate and refer to project lead for possible replacement

		have technical issues			
	Complaints raised by project staff	Unfair decisions related to pay or disciplinary actions	Gather information and refer to project lead	Project manager	5

2.1.5 Public Comments

The project activity was listed for public comments on the Verra platform on **14/07/2023** and completed the public commenting period of 30-days on **13/08/2023**. During which, there were no public comments.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

TEM has been developing carbon projects since 2015 and has its expertise in managing and registering carbon projects, offering its clients comprehensive end-to-end solutions to traversing the carbon market landscape. TEM has currently 6 registered Australian Carbon Credit Unit (ACCU) registered projects.

Where additional expertise is required for the project activity, TEM consults with in-country experts and universities to acquire a clearer understanding of the socio-economic situation on the field within which the project activity is residing.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risks identified	The wellbeing of the end-users are improved by less smoke and more safe cooking devices compared to the baseline cooking practice over open fire cooking.
Risks to stakeholder participation	Low adoption or participation rates of end-users in the project	Participation in the project is voluntary and thus, no household is forced to be involved. Outreach and engagement to educate the end users on the direct and indirect benefits of improved cookstoves over open fire cooking, and continuous awareness campaigns to boost adoption rates.
Working conditions	Safe working conditions for support staff employed in awareness and distribution of cookstoves	Support of local police, private security teams and the involvement of community leaders in cookstove distribution planning enables the

		mitigation of identified security risks for on-ground teams.
Safety of women and girls	No risk identified	The project activity increases women's and girls' safety by reducing the number of trips to the forest to collect firewood through the adoption of ICS that are safer than open fire practices.
Safety of minority and marginalized groups, including children	No risk identified	The ICS distribution and monitoring do not affect the safety of minority and marginalized groups. The improved cookstoves reduce the risks associated with open fire cooking practices for children.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	End-of-life waste from cookstoves disposal	The ICS is made of stainless-steel parts that are recyclable and do not contribute to hazardous waste. In addition, the project activity decreases the risk of pollutants (smoke from cooking) and does not produce any form of pollution.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

In accordance with PNG's laws and regulations, there are acts that has been adopted in this project to ensure there is respect for human rights and equity. Some examples that are followed are The Employment Act No. 54 of 1978, The Child Welfare Act 1961, The Discriminatory Practices Act 1963, and The Workers Compensation Act 1978. These laws are further described in our PDD.

In this project, any form of discrimination and harassment is opposed fully and will not be tolerated. Additionally, any forced labour, human trafficking or child labour is opposed in this project and serious action will be taken to the parties involved.

This is stated in our TEM Employee Handbook and Modern Slavery Policy:

	Risks identified[18]	Mitigation or preventative measure(s) taken
Discrimination	Unfair treatment due to a certain attribute such as a worker's racial or ethnic background, political or religious beliefs, stereotypes, age, race or disability.	Under the 'Discrimination' section of the TEM Employee Handbook, any discrimination faced is reportable and action will be taken against the perpetrator. Severe or repeated breaches can lead to formal discipline up to and including dismissal.

[18] The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Sexual harassment	Sexual Harassment may affect a person mentally and result in a negative change in behaviour.	Under Section 13 of the TEM Employee Handbook, all sexual harassment faced is reportable and the policy provides the appropriate personnel to contact in the event of a sexual harassment. Serious actions will be taken against the perpetuator if found guilty.
Gender equity in labor and work	Workers being treated unfairly due to gender, causing a divide in the workplace.	Workers will be paid fairly based on only their qualifications and their merits.
Forced labor	Workers being treated unfairly and working involuntarily.	Under TEM's Modern Slavery Policy, TEM is dedicated to preventing modern slavery in all its forms, including forced labour, human trafficking, child labour, and debt bondage.
Child labor	Children under 18 are employed, which affects their education and mental and physical health.	Under TEM's Modern Slavery Policy, TEM is dedicated to preventing modern slavery in all its forms, including forced labour, human trafficking, child labour, and debt bondage.
Human trafficking	People being transported around to benefit from their work or service as a form of forced labour.	Under TEM's Modern Slavery Policy, TEM is dedicated to preventing modern slavery in all its forms, including forced labour, human trafficking, child labour, and debt bondage.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No Risk Identified	This project is a voluntary project activity that involves the distribution of ICSs to individual households free of charge.

2.3.3 Indigenous Peoples and Cultural Heritage

Local people from the project area have been employed by TEM PNG Cookstoves Project Pte. Ltd to run the day-to-day operations of the project activity.

The project is aligned and strictly complies with international laws and regulations such as the UN Declaration on the Rights of Indigenous People Rights. Any form of opposition to these laws will not be tolerated. Risks identified were based on a lack of knowledge on Indigenous Peoples and Cultural Heritage in the project area and their respective preventive measures have been included in the table below:

Risks identified	Mitigation(s) or preventative measure taken

Lack of knowledge of the rights of the Indigenous Peoples	Under the regulatory surplus of the project activity, this project will follow and comply with the UN Declaration on the Rights of Indigenous Peoples. Personnel involved in this project will be well educated on the rights of the Indigenous people. Any issues or problems observed can be brought up as a Grievance as explained more in section 2.1.4.
Lack of recognizing, respecting and promoting rights of the Indigenous Peoples	Under the regulatory surplus of the project activity, this project will aim to follow both international rights such as UN Declaration on the Rights of Indigenous Peoples Rights and National rights such as the Land Act 1996 and Land Group Incorporation Act 1974. These rights will be strictly followed and implemented to ensure that the Indigenous People have their rights and cultural heritage protected. Any issues or problems observed can be brought up as a Grievance as explained more in section 2.1.4.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No Risk Identified	This project is a voluntary project activity that involves the distribution of ICSs to individual households free of charge.

2.3.5 Benefit Sharing

Summary of the benefit sharing plan	Any household within the project boundary is eligible to be part of the project as long as they are effectively onboarded into the project activity first through the Household Registration Process and Cookstove Distribution Process. By completing these two steps, the end-user's household will receive an ICS free of charge and reap the direct and indirect benefits of using an improved cookstove.
Benefit sharing during the monitoring period	The Household Registration Process and Distribution Process occurred between 23 September 2024 to 31 October 2024 during which 19,206 ICS were distributed to

	unique households. Through the use of the ICS, end-users reaped the direct and indirect benefits of the ICS such as: • Reducing the amount of money and time spent of buying or collecting firewood • Improvement of air-quality within the kitchen/cooking dwelling due to efficient burning of biomass • Reduction of forest degradation in the surrounding area due to the reduced need for biomass for cooking or heating For a full list of benefits, please refer to Section 1.12 of this document.

2.4 Ecosystem Health

	Risk identified	Mitigation or preventative measure(s) taken during the monitoring period
Impacts on biodiversity and ecosystems	No Risks Identified	The technology of the project involves the reduction of biomass for cooking or heating and will not impact the biodiversity and ecosystems in any way through soil degradation, erosion, and additional water consumption and stress.
Soil degradation and soil erosion		
Water consumption and stress		

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

Species or habitat	*Not applicable*
Areas needed for habitat connectivity	*Not applicable*

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	*Not applicable*	*Not applicable*
Areas for habitat connectivity	*Not applicable*	*Not applicable*

2.4.2 Introduction of Species

The technology of the project involves the reduction of biomass for cooking or heating through Improved Cookstove technology. It does not introduce any species of animal or plants into the project area, as such this section is not applicable.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
Not applicable	*Not applicable*	*Not applicable*	*Not applicable*

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
Not applicable	*Not applicable*

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	*Not applicable*	*Not applicable*

2.4.3 Ecosystem Conversion

The technology of the project involves the reduction of biomass for cooking or heating through Improved Cookstove technology. It does not conduct any ecosystem conversion within the project area; hence this section is not applicable.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	*Not applicable*	*Not applicable*

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0050	VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves	1.0
Standard	-	Sampling and surveys for CDM project activities and programmes of activities	9.0
Tool	30	Calculation of the fraction of non-renewable biomass	4.0
Guidelines	-	Sampling and surveys for CDM project activities and programmes of activities	4.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VM0050	The project activity corresponds to: a) Replacement of non-renewable biomass (e.g., firewood, charcoal)-fired cookstoves with any of the following: i) More efficient project devices that use the same fuel as in the baseline; ii) Efficient project devices fired by renewable biomass or bioethanol; iii) Efficient project devices fired by liquefied petroleum gas (LPG); or iv) Electric-powered project devices.	**Applicable** The project activity involves the replacement of non-renewable biomass with more efficient project devices that use the same fuel as in the baseline (firewood), therefore this criterion is met.

	b) Replacement of solid or liquid fossil fuel (e.g., coal, kerosene)-fired cookstoves with any of the following: i) Efficient project devices fired by renewable biomass or bioethanol; ii) Efficient project devices fired by LPG; or iii) Electric-powered project devices.	
VM0050	Project devices are used in households, communities, institutions, or small or medium enterprises (SMEs), collectively referred to in this methodology as the "target population."	**Applicable** The project activity will be used in an end-user's household for heating and cooking purposes. Therefore, this criterion is met.
VM0050	Where renewable biomass is used, it is exclusively renewable and qualifies as one of the following: a) A by-product, residue, or waste stream from agriculture, forestry, and related industries; or b) Originating from dedicated plantations that comply with all relevant applicability conditions in the most recent version of CDM TOOL16.	**Does not Apply** The project activity does not use renewable biomass therefore this criterion does not apply.
VM0050	Where biomass residues are used, they would have been left to decay or burned without energy recovery before implementation of the project activity, and their use does not involve a decrease in carbon pools – in particular of dead wood, litter, or soil organic carbon – on the land areas from which the biomass residues originate.	**Does not Apply** The project activity does not use biomass residues therefore this criterion does not apply.

VM0050	Where biomass residues from a production process are used, project implementation does not result in an increase in the processing capacity of raw input or any other substantial changes (e.g., product change) in this process.	**Does not Apply** The project activity does not use biomass residues therefore this criterion does not apply.
VM0050	Where more than one type of renewable biomass is used, each of the biomass types used complies with the applicability conditions.	**Does not Apply** The project activity does not use renewable biomass therefore this criterion does not apply.
VM0050	Where project activities introduce renewable biomass such as charcoal, it is renewable charcoal produced by efficient charcoal production processes (e.g., retort sedentary kilns, improved sedentary kilns, Casamance kilns). Methane produced during the charcoaling process is captured and destroyed or combusted for energy purposes.	**Does not Apply** The project activity does not introduce renewable biomass such as charcoal into the project boundary therefore this criterion does not apply.
VM0050	Project devices using renewable biomass (fuel-switch) or non-renewable biomass (improved efficiency) are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 25%.	**Applicable** The project activity uses non-renewable biomass single-pot portable cookstoves that has an initial thermal efficiency of 52%, therefore this criterion is met.
VM0050	Project devices using LPG or bioethanol are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 30%.	**Does not Apply** The project activity uses ICSs that utilize firewood as a fuel and does not involve LPG or bioethanol therefore this criterion does not apply.
VM0050	Electric project devices meet the maximum risk factor score of 15 on the	**Does not Apply** The project activity uses ICSs that utilize firewood as a fuel

	Cookstove Durability Protocol and have the following minimum thermal efficiency: a) Hot plates and electric hobs: 40% b) Induction stoves and other electric stoves: 70%	for cooking and heating purposes, therefore this criterion does not apply.
VM0050	Project devices using LPG comply with all of the following conditions: a) The baseline fuel either includes non-renewable biomass or is a more carbon-intensive fossil fuel. b) The project has a provision for metering LPG supplied to each consumer at the LPG filling station, in order to determine household LPG consumption; and c) The project does not seek to issue any carbon credits for periods after 31 December 2045.	**Does not Apply** The project activity uses ICS that utilize firewood as a fuel for cooking and heating purposes, therefore this criterion does not apply.
VM0050	Electric project devices use the following electricity sources: a) Decentralized renewable energy systems: Decentralized energy systems using fossil fuels are not eligible, except for backup generators that supply less than 1% of the annual electricity of the decentralized renewable energy system. b) Self-generated renewable electricity (with a maximum of 20% electricity from non-renewable sources for backup); or c) National or regional electricity grid.	**Does not Apply** The project activity uses ICSs that utilize firewood as a fuel for cooking and heating purposes, therefore this criterion does not apply.
VM0050	The project proponent designs incentive mechanisms to reduce the use of	**Applicable**

	inefficient baseline devices and practices that can be replaced by the project devices and describes these mechanisms in the project description.	The project proponent employs the following incentive methods to reduce the use of inefficient baseline devices and practices: <u>Household Education and Encouragement:</u> The project proponent, through cookstove ambassadors, routinely visits end-users throughout the project lifetime to collect feedback, maintain the project cookstoves, and address questions or queries regarding the project activity to encourage the continued use of the project cookstoves. <u>Behavioural Change Campaigns:</u> Awareness champaigns are also held by field assistants and cookstove ambassadors in the form of community meetings featuring cooking demonstrations, peer testimonials within villages to promote and educate households on the health, environmental, and economic benefits of transitioning to improved cookstoves, thereby motivating voluntary behaviour change. Additionally, the project proponent works with the Western Pacific University (WPU) of Papua New Guinea to develop and train locals on

		project management and carbon knowledge. Graduates of this course would be able to work with the project proponent formally during the project lifetime. <u>Monitoring and Enforcement:</u> During the project lifetime, visits done for household education and encouragement will also monitor the usage rate of the ICS. This effectiveness of the abovementioned incentive mechanisms will be monitored as a decrease in stove stacking factor. <u>Additional Incentive Engagement:</u> From the results of the abovementioned monitoring, villages showing the most progress will be rewarded with celebratory 'Cook-Off' Events featuring demonstrations, knowledge-sharing, and success stories. Testimonials will be gathered for short-form videos to boost awareness via social media, while non-adopters will be consulted to identify and address adoption barriers for future training improvements. Hence, this criterion is met.
VM0050	Where a project device has ended its technical life, the project proponent either replaces it with a comparable or better project device or retrofits its essential	**Applicable** Where the ICS has ended its technical life, the project proponent does not replace the

	components to continue meeting the minimum service level requirements (i.e., thermal energy generation), otherwise no further emission reductions may be claimed for the project device.	ICS. Instead, no further emission reductions will be claimed for the ICS, therefore this criterion is met.
VM0050	Project proponents implement a method for the distribution and identification of project devices that avoids double counting of emission reductions by other mitigation actions and includes unique product identification on the stove itself at the time of distribution/sale (e.g., program logo, alpha/numeric ID, and end-user location, such as geographic coordinates, complete address information).	**Applicable** In order to avoid double counting emission reductions through the method of distribution, each ICS will carry a USN and the distribution of each device is logged into an electronic database which includes unique information for each recipient (i.e., Unique Serial Number, geographic coordinates, etc.) upon receipt of the device. The electronic database combined with internal checks done by the PP will ensure that no USNs are double counted. During the monitoring period, survey samples will be only taken from the USNs that are in the electronic database of end-users eliminating the risk of double counting from any other projects within the same geographical location.
VM0050	The project complies with any national, sub-national, or local regulations or guidance for the installation, commercialization, distribution, and use of improved cookstoves and/or fuel supply and use for the target population. National, regional, and local regulatory frameworks for the provision of the type of thermal energy services provided by the project activity must be documented.	**Applicable** The project activity complies with any national, sub-national, or local regulations or guidance for the installation, commercialization, distribution, and use of improved cookstoves and/or fuel supply and use for the target population. At every stage of project inception, the project proponent has been in touch with all levels of relevant

		authorities and have gained approval, therefore this criterion is met.
VM0050	Where project activities reduce emissions from non-renewable biomass, including firewood and charcoal, the risk of double counting is assessed on a national basis by evaluating at validation and crediting period renewal whether there are REDD+ projects or jurisdictional REDD+ programs whose project boundary overlaps with the expected fuel source area of the project. The project proponent must report on the findings of this assessment for informational purposes in the project description.	**Applicable** Currently, there are no REDD+ projects or jurisdictional REDD+ programs with the boundary of PNG Highlands[19], therefore this criterion is applicable.

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Thermal energy generation	CO_2	Yes	Major source
		CH_4	Yes	May be significant for some fuels
		N_2O	Yes	May be significant for some fuels
		Other	-	-
Project	Thermal energy generation	CO_2	Yes	Major source
		CH_4	Yes	May be significant for some fuels
		N_2O	Yes	May be significant for some fuels
		Other	-	-

The baseline and project boundaries are presented in Figure 11.

[19] This information has been cross-checked with the Verra Registry.

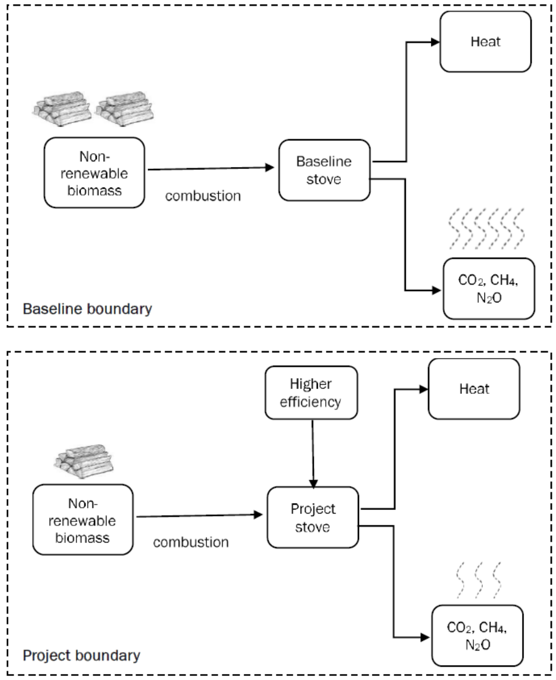


Figure 11: Baseline and Project Scenarios

Within the project boundary, less biomass burning gases and particulates are emitted.

3.4 Baseline Scenario

The project activity was formally registered on 14/06/2024 under the methodology VMR0006: Energy Efficiency and Fuel Switch Measures in Thermal Applications, v1.2. and underwent a methodology update to VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0 as per Verra Guidelines[20].

In accordance with the *Procedure to Change Methodology through a Project Description Deviation v4.0*, the project is not required to undergo a baseline reassessment, as its technology does not fall under any of the categories listed in Section 2.1.3, item 1, footnote 3. Instead, as outlined in the procedure for projects not subject to baseline reassessment, TEM has applied the new methodology to determine the most plausible baseline scenario and to calculate baseline emissions for the remainder of the crediting period, in line with the revised baseline scenario requirements.

As per the VM0050 v1.0, Section 6.1, the selection and justification of the baseline scenario is completed in the following steps.

Step 1: Identify alternative baseline scenarios

The baseline scenario is the continued use of non-renewable wood fuel (firewood/charcoal) or fossil fuel (coal/kerosene) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

a) A third-party consultant team contracted by TEM conducted a baseline study survey and measurement campaign which concluded in November 2022 (file titled: TEM

[20] VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0, https://verra.org/methodologies/vm0050-energy-efficiency-and-fuel-switch-measures-in-cookstoves-v1-0/

2025_Cookstoves Highlands region report_v3.pdf) with the results included in a report that was audited by an esteemed in-country forester (file titled: SPFEC Letter to TEM-Review of Cookstoves Highlands region report_17-01-2025.pdf). During which it was found that Papua New Guinea households consumed significant volumes of firewood, with reports suggesting that the consumption of up to 10 - 13 million tonnes of firewood per year. Among others, household cooking was responsible for the greatest share of energy demand as most people in the country still use traditional open fire cooking methods in both rural and urban settings. Such dependency on firewood resources not only affects the forests, but also the welfare, finance, and labour of the local families.

b) The questions administered in the baseline survey[21] conducted is fully aligned with the binding baseline questionnaire outlined in Appendix 3 of VM0050 v1.0, with the exception of the questions on household profile, where a suitable deviation was applied to approximate the average household size with an error of <1% (please see methodology deviation in Section 3.6). The structure and content of the survey meets the methodological requirements, ensuring consistency with the standardized data collection approach. In addition to meeting the questionnaire requirements, baseline fuel consumption values were also directly measured in accordance with the methodology's guidance, ensuring that both qualitative and quantitative data necessary for establishing a credible and compliant baseline scenario were captured accurately.

Typical rural households in the highland's region of Papua New Guinea cook their food using the traditional open fire method while some use open drum stoves which is still considered as traditional method:

Figure 12: Traditional cooking practices in the project area

As per the picture above, picture A. depicts cooking using Open Drum Stoves (ODS) while B. depicts Open Fire Cooking (OFC). Traditional OFC has been practiced for decades, and it is still the main method for cooking to date. The ODS were introduced into the PNG society approximately 30 years ago to keep the aluminium pots clean from direct heat from carbon dioxide released from the fire and they are currently considered a traditional cooking method.

[21] Original baseline survey has been submitted to the VVB for validation and verification.

In the baseline survey, questions were structured to the type of cookstove that is currently being utilized in each surveyed household and the period that cookstove was used, The results of the survey showed that over 81.00% of the households in the Western Highland Province use traditional OFC and or ODS for cooking their meals while in Southern Highlands Province , almost 99.00% of the household rely entirely on traditional OFC and ODS for cooking. Collectively, just about 90.00% of the surveyed households use traditional OFC/ODS in the baseline study areas.

Household Selection Criteria

In addition to proving the baseline above, the project proponent will also include additional safeguards to ensure that this baseline is maintained through the lifetime of the project activity through the implementation of a 'Household Selection Criteria'.

The objective of the household selection criteria is to guarantee that households that are receiving the ICSs use firewood as their primary method of cooking, hence being applicable under the baseline scenario. The selection criteria will take the form of a questionnaire at the start of the ICS distribution, and will include binomial questions (yes/no answers) to assess the household's primary method of cooking fuel as follows:

1. Do you cook any meals in the house without using firewood?
2. Do you use charcoal for cooking in your household?
3. Do you use electric cooking in your household?
4. Do you use LPG for cooking in your household?

If any of the responses to these questions indicate that firewood is not the main fuel used in the household for cooking, this household will not be selected to participate in the project activity. These questions may be amended and improved upon with each ICSs distribution cycle.

Established Controlled Households

As per VM0050 v1.0 Section 6.2 and Correction 1, follow-up baseline surveys must be conducted every 5-years in controlled households and do not participate in the project. These control households must be established prior to validation and must be shown to be statistically equivalent to the pre-project conditions of project households regarding baseline fuel type(s) and percentage of use by the target population, source(s) of each baseline fuel, and baseline technologies for cooking.

The control houses are identified, however not registered on a project database, due to practical on-the-ground operational concerns that can cause risks to the longevity of the project. House registrations trigger an anticipation of cookstoves to be distributed imminently, while by design we are not distributing cookstoves in these households. Over time, this expectation can escalate into frustrations and can affect the safety and harmony of the field teams working in the area and other neighbouring houses where project devices were distributed.

Therefore, it is not practical to register all the households on a project database, and instead an entire ward, which is the smallest administrative district in PNG, is chosen as a control. The control households are identified to be in the Koalilombo ward[22] of the Southern Highlands Province within the project boundary. Based on PNG Census 2011[23], this ward has an estimated 341 households and a population of approximately 1,992 people. A KML file demarcating the geographical extend of Koalilombo ward is provided to the VVB to validate that the area falls

[22] GPS Coordinates: 6º 25' 56.7" S 143º 54' 11.2" E, https://maps.app.goo.gl/6J9omZzdJf47jdHL8

[23] National Statistics Office, Papua New Guinea National Population & Housing Census 2011, 'Count me in' (2011)

under the project boundary. This area has the same baseline and is within the project boundary, and is in close proximity to the cookstove distribution areas, and therefore a suitable control area.

TEM will ensure that no ICSs will be distributed[24] to households within this ward throughout the lifetime of the project activity.

Step 2: Consider existing and forthcoming government policies and legal requirements

The baseline identified above is not inconsistent with any existing or forthcoming laws, regulations or government policies, as there are no known initiatives that focuses on replacing traditional cooking practices. The country of the proposed project activity does not have any minimum efficiency standards or air quality requirements that are applicable for traditional cooking practices like OFC/ODS.

Step 3: Assess financial, institutional, and information barriers

Table 7: Assessing barriers to implementation

Financial barriers	No similar activities related to improving cooking efficiency have been implemented in the provinces under the project area. The cost to transition to more efficient forms of cooking are higher than what the end-users are able to afford. There are no known grants or aid from domestic or international organisations available for clean cooking projects in the region that might prevent the identified baseline scenario from occurring.
Institutional barriers	There are no known investors or corporate structures that are incentivised to reduce domestic energy consumption or improve energy efficiency in the provinces under the project area, that might prevent the baseline scenario identified.
Information barriers	There is a lack of awareness of financial and non-financial benefits of ICS devices that require to foster active awareness campaigns and engagement initiatives to boost adoption rates and behavioural change.

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☒ Annex 1 country ☐ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

[24] A KML file demarcating the geographical extend of Koalilombo ward is provided to the VVB to validate that no cookstoves are distributed in the region during each monitoring period.

☐ Yes ☐ No

As Papua New Guinea is included in Annex 1 of the UNFCCC, the above point is not applicable to the project additionality.

3.5.2 Additionality Methods

The methodology VM0050 v1.0 uses activity and project methods for the demonstration of additionality.

Activity Method

Step 1: Regulatory Surplus

There is no mandated government programme or policy in the host country of this project ensuring distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or regulatory framework, or for the UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework.

Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by TEM.

Step 2: Positive List

The applicability conditions of this methodology represent the positive list. As described in Section 3.2, the project meets all these conditions. The project also meets the following conditions as per VM0050 v1.0 wherein:

1. The project activity introduces efficient biomass-fired cookstoves that replace inefficient biomass-fired cookstoves.
2. The project distributes cookstoves at zero cost[25] to the end-user and has no other source of revenue other than the sale of GHG credits.
3. Project activities are not implemented as part of government schemes nor is the project supported by multilateral funds[26].

Step 3: Project Method

As the project activity meets the requirements of the positive list, there is no need to apply the investment analysis as stated in the methodology.

The project can therefore be deemed additional.

[25] This is justified through end user agreements and validated by Earthood during the remote site visit.

[26] This is justified through TEM's Funding Letter.

3.6 Methodology Deviations

The age distribution captured during baseline survey in October-November 2022 for the baseline of VMR0006, differs slightly from the binding questionnaire provided in Appendix 3 of VM0050 v1.0. This data is utilised to calculate the adult equivalent per household, which is in turn used to estimate the ex-ante parameter *Average Household Size (Hhi)*.

The data collected from the baseline survey in 2022 is used with a reasonable approximation to calculate the average adult equivalent as explained in Table 07.

Household profile (VM0050 binding questionnaire)	Value to be used for calculating Adult Equivalent	Household profile (baseline survey)	Value used for calculating Adult Equivalent
Females, over 14 years	0.8	**Adults**, 16 -63 years	0.9[27]
Males, 15-59 years	1.0		
Males, over 59 years	0.8	**Adults**, 64 years and above	0.8
Children, 0-14 years	0.5	**Children**, under 15 years	0.5

Section 3.20 of VCS Standard v4.7, notes that "projects are permitted to deviate from the procedures set out in methodologies in certain cases, such as where alternative methods may be more efficient for project-specific circumstances, or where the deviation will achieve the same level of accuracy or is more conservative than what is set out in the methodology."

As the variation between the two approaches is <1%, demonstrated with recent field measurements to the auditor, this is considered a project-specific circumstance where the alternate method proposed is more efficient than repeating the baseline survey, while achieving the same level of accuracy and is an acceptable methodology deviation.

[27] A simple average of 1 and 0.8 is used to approximate for an adult assuming 50:50 distribution of male and female, as the male-to-female sex ratio in PNG is ~1 from the 2021 PNG census (https://www.nso.gov.pg/statistics/population/)

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline emissions are calculated as follows:

$$BE_y = \sum_{i,j,k} EC_{i,y} \times N_{j,k,y} \times n_{j,k,y} \times \left(EF_{b,i,CO2} \times f_{NRB,y} + EF_{b,i,nonCO2}\right) \quad (1)$$

Where:

BE_y	=	Baseline emissions during year y (t CO_2e)
$EC_{i,y}$	=	Average energy consumption of baseline device type i in year y (TJ)
$N_{j,k,y}$	=	Number of commissioned project devices of type j from batch k in year y
$n_{j,k,y}$	=	Proportion of commissioned project devices of type j from batch k that remain operating in year y (fraction)
$EF_{b,i,CO2}$	=	CO_2 emission factor for fuel used by baseline device type i in the baseline scenario (t CO_2/TJ)
$f_{NRB,y}$	=	Fraction of woody biomass that is established to be non-renewable used by baseline device in year y; this variable is not considered for fossil fuels (fraction)
$EF_{b,i,nonCO2}$	=	Non-CO_2 emission factor for fuel used by baseline device type i in the baseline scenario (t CO_2e/TJ)
i	=	Baseline device type and its respective fuel type
j	=	Project device type and its respective fuel type

Average Energy Consumption of Baseline Device ($EC_{i,y}$)

The average energy consumption of baseline device type i is calculated as follows:

$$EC_{i,y} = BC_{b,i,y} \times NCV_{b,i} \quad (2)$$

Where:

$BC_{b,i,y}$	=	Fuel used per baseline device type i during year y (tonnes)[17]
$NCV_{b,i}$	=	Net calorific value of baseline fuel for baseline device type i (TJ/tonne)

The quantity of fuel that would be used in the baseline scenario was determined through Option 1: Measurement Campaign conducted by Simi Oa Consultancy Services that is a third-party consultant team contracted by TEM along with the baseline study survey concluded in November 2022. A total of 220 houses were selected for the baseline study with 110 houses each from the Southern Highlands Province and Western Highlands Province. The selected houses were provided with an individual code from 001 to 220. Of the total number of houses, 160 were used for control cooking tests (CCT) comparing firewood consumption using improved cookstoves (ICS, n=80) with traditional open fire cooking (OFC, n=80). At the end of the campaign, sampling challenges enabled the collection of firewood consumption data on 131 households in both provinces, of which 60 houses used ICS and the remaining 71 houses used traditional OFC. The results of the measurement campaign yielded a value of 5.97 tons as the firewood usage per year on a baseline stove (OFC) (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf).

As the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 71 samples and achieved a precision value 12.36%, The lower bound value of 5.24 tons/year is used after applying a conservativeness deduction following Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0.

Table 8: Conservativeness deduction for Fuel consumption

	Baseline (OFC)	Project (ICS)
Avg adult consumption/year (tons)	**1.23**	**0.28**
Standard Deviation	0.77	0.17
Precision	12.36%	12.98%
Avg Household size	**4.85**	
Avg consumption/ household /year (tons)	5.97	1.36
Upper bound	6.71	**1.54**
Lower bound	**5.24**	1.18
After conservativeness correction		
Fuel consumption/household/year	**5.24** (BC_b)	**1.54** (BC_p)

Cross-check of $EC_{i,y}$ to address cookstove stacking:

Stove stacking will be addressed by comparing the quantity of baseline energy consumption determined by both options above ($EC_{i,y}$) to energy used in the project scenario ($EC_{p,y}$) using back-calculation. Where the results indicate that baseline consumption ($EC_{i,y}$) is higher than that indicated by back calculation from the project scenario ($EC_{est,y}$) then stove stacking is occurring. The back-calculation results ($EC_{est,y}$) must be applied in Equation (1) as a conservative cap and is calculated and defined as follows:

$$EC_{est,y} = EC_{p,y} \times \frac{\eta_{new,avg,y}}{\eta_{old,avg}} \quad (3)$$

Where:

$EC_{est,y}$	=	Back-calculated energy consumption of the potential mix of devices and fuels in the baseline in year *y* (TJ)
$EC_{p,y}$	=	Energy used in project scenario by project devices during year *y* (TJ)
$\eta_{new,avg,,y}$	=	Weighted average efficiency of project devices in year *y* (fraction)
$\eta_{old,avg}$	=	Weighted average efficiency of baseline devices that are eplaced by project devices (fraction)

$EC_{p,y}$ must be determined as follows:

$$EC_{p,,y} = \sum_{j,k} BC_{p,j,k,y} \times NCV_{p,j} \quad (4)$$

Where:

$BC_{p,j,k,y}$	=	Average quantity of fuel used by project device type *j* from batch *k* during year *y* (tonnes or m^3)
$NCV_{p,j}$	=	Net calorific value of project fuel used in project device type *j* (TJ/tonne or TJ/m^3)

<u>Baseline Emission for Project Instance:</u>

For ex-ante calculation purpose, the assumption below is applied for first project activity instance:

- The first Project Instance installs 100,000 ICSs in the first year throughout the crediting period.
- The end date of installation of project activity instance is considered as operational date of that project activity instance.
- Based on manufacturer specifications (Table 2) the life span of each individual ICS is 10 years; thus, the operational lifetime of each project activity instance is taken as 10 years.
- Annual cookstove loss rate is estimated at 5% for the initial ER calculations. The value is a conservative estimate although there were no losses observed during surveys. This will be a monitored parameter for the actual ER calculations during verification.
- $BC_{b,i,y}$ is fixed as 5.24 tonnes/year as per TEM's Baseline Study after applying conservativeness deduction for 90/10 confidence and precision as per Table 8 (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf) showed.
- For ex-ante calculations, it is assumed that no cookstove stacking is taking place, therefore ECi,y is equals to $EC_{p,y}$. Please refer to ER Calculation sheet for full details.

- $f_{NRB,b,y}$ is fixed as 0.81 as per TEM's literature review and baseline calculations.
  - The fNRB value is calculated using CDM Tool 30 version 4.0, using publicly available national statistics data reported by the PNGFA, including the forest regrowth rates from Biennial Updates made to UNFCCC in 2022. The data inputs and the sources are explained in the relevant table in Section 5.1.
  - Comparison with data from Bailis et.al. (2015): The value for fNRB modelled and predicted by Bailis et. al. (2015) in PNG ranges from 0.20 to 1 for different provinces with a country average of 0.40. This study modelled the fNRB numbers using data on woodfuel demand and supply from 2009 global datasets and regrowth rates, correlated with 2,800 spatial observations, which were extrapolated for other geographies. It is unclear how many of these locations, if any, were located in PNG. The woodfuel data for PNG was obtained from the FAOSTAT global dataset for Forestry Production and Trade (2009).
  - All the inputs used in the CDM Tool 30 calculation are from PNG national statistics reported by the PNGFA (2019) including the forest regrowth rates from Biennial Updates made to UNFCCC in 2022. These localised country specific inputs provide more confidence on the fNRB calculations than pixel-based models, as they are significantly recent than the inputs used by Bailis et. al and are independently verified by an in-county forestry expert.
  - The project has increased the level of integrity in the reporting of GHG ERs through the following steps: 1) By working with local PNG forestry experts to accurately estimate the fNRB for the country, reaching to a conservative value of 0.81 (file titled: TEM (2023). f_{NRB} calculation_Final and audited report titled: SPFEC Letter to TEM-Review of FnRB Calculations.pdf; and 2) in alignment with new methodology requirements, additional 26% adjustments were made using the μ_d parameter in our ERs calculations. Overall, the proposed approach leads to an effective fNRB value of 0.55 (= 0.81- 0.26) bringing rigour into our estimates by reducing a potential 40% of over crediting that would have taken placed if the Old United Nations (UN) default value of fNRB (0.99) was applied.

Determining Average Energy Consumption of Baseline Device (ECi,y)

Table 9: Intermediate ER Calculations (Equation 2)

Equation 2	$EC_{i,y} = BC_{b,i,y} \times NCV_{b,i}$		
	$BC_{b,i,y}$	$NCV_{b,i}$	$EC_{i,y}$
	5.24	0.02	0.08

Determination of Baseline Emissions (BE_y)

Applying the Annual cookstove loss rate is of 5%, the ex-ante number of cookstoves over the project lifetime is presented below:

Table 10: Loss rate of ICS each year

	$N_{j,k,y}$
Year 1	100,000
Year 2	95,000
Year 3	90,250
Year 4	85,738
Year 5	81,451
Year 6	77,378
Year 7	73,509
Year 8	69,834
Year 9	66,342
Year 10	63,025

Therefore, baseline emissions will be calculated as follows:

Table 11: Intermediate ER Calculations (Equation 1)

Equation 1	$BE_y = EC_{i,y} \times N_{j,k,y} \times n_{j,k,y} \times (EF_{b,i,CO2} \times f_{NRB,y} + EF_{b,i,nonCO2})$						
	$EC_{i,y}$	$N_{j,k,y}$	$n_{j,k,y}$	$EF_{b,i,CO2}$	$f_{NRB,y}$	$EF_{b,i,nonCO2}$	BE_y
Year 1	0.08	100,000	0.90	112	0.60	9.46	**563,108**
Year 2	0.08	95,000	0.90	112	0.60	9.46	**534,953**
Year 3	0.08	90,250	0.90	112	0.60	9.46	**508,205**
Year 4	0.08	85,738	0.90	112	0.60	9.46	**482,795**
Year 5	0.08	81,451	0.90	112	0.60	9.46	**458,655**
Year 6	0.08	77,378	0.90	112	0.60	9.46	**435,723**
Year 7	0.08	73,509	0.90	112	0.60	9.46	**413,936**
Year 8	0.08	69,834	0.90	112	0.60	9.46	**393,240**
Year 9	0.08	66,342	0.90	112	0.60	9.46	**373,578**
Year 10	0.08	63,025	0.90	112	0.60	9.46	**354,899**

4.2 Project Emissions

Project emissions are calculated as follows:

$$PE_y = PE_{energy,y} + PE_{others,y} \tag{6}$$

Where:

PE_y	=	Project emissions during year *y* (t CO_2e)
$PE_{energy,y}$	=	Project emissions from energy consumption of project devices in year *y* (t CO_2e)
$PE_{others,y}$	=	Project emissions from other sources in year *y* (t CO_2e)

$PE_{energy,y}$ from Biomass is determined as follows:

$$PE_{energy,y} = \sum_j \sum_k BC_{p,j,k,y} \times N_{j,k,y} \times NCV_{p,j} \times n_{j,k,y} \times \left(EF_{p,j,CO2} \times f_{NRB,y} + EF_{p,j,nonCO2}\right) \tag{7}$$

Where:

$EF_{p,j,CO2}$	=	CO_2 emission factor for fuel used by project device type *j* in the project scenario (t CO_2/TJ)
$EF_{p,j,nonCO2}$	=	Non-CO_2 emission factor for fuel used by project device type *j* (t CO_2e/TJ)

$BC_{p,j,k,y}$ is estimated using Option 1: Kitchen Performance Test wherein a measurement campaign was conducted by a third-party consultant team contracted by TEM along with the baseline study survey concluded in November 2022. A total of 220 houses were selected for the baseline study with 110 houses each from the Southern Highlands Province and Western Highlands province. The selected houses were provided with an individual code from 001 to 220. Of the total number of houses, 160 were used for control cooking tests (CCT) comparing firewood consumption using improved cookstoves (ICS, n=80) with traditional open fire cooking (OFC, n=80). At the end of the campaign, sampling challenges enabled the collection of firewood consumption data on 131 households in both provinces, of which 60 houses used ICS and the remaining 71 houses used traditional OFC. The results of the measurement campaign yielded a value of 1.36 tons as the firewood usage per year on the project stove (ICS) (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf).

As the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 60 samples and achieved a precision value 12.98%, the upper bound value of 1.54 tons/year is used after applying a conservativeness deduction as per Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0.

Table 12: Conservativeness deduction for Firewood consumption

	Baseline (OFC)	Project (ICS)
Avg adult consumption/year (tons)	**1.23**	**0.28**
Standard Deviation	0.77	0.17
Precision	12.36%	12.98%
Avg Household size	**4.85**	
Avg consumption/ household /year (tons)	5.97	1.36
Upper bound	6.71	**1.54**
Lower bound	**5.24**	1.18
After conservativeness correction		
Fuel consumption/household/year	**5.24** (BC_b)	**1.54** (BC_p)

$PE_{others,y}$ is determined as follows:

$$PE_{others,y} = PE_{transp,y} + PE_{prod,y} + PE_{fugitive,y} + PE_{backup,y} \quad (9)$$

Where:

$PE_{transp,y}$ = Project emissions due to fuel transportation in year *y* (t CO_2e)

$PE_{prod,y}$ = Project emissions due to fuel production in year *y* (t CO_2e)

$PE_{fugitive,y}$ = Fugitive emissions in year *y* (t CO_2e)

$PE_{backup,y}$ = Project emissions from backup generators in year *y* (t CO_2e)

As the project design and boundary does not involve fuel transport, production, fugitive emissions or emissions from backup generators, this calculated parameter does not apply to the project activity.

Project Emission for Project Instance:

For ex-ante calculation purpose, the assumption below (in addition to the assumptions stated in Section 4.2) is applied for first project activity instance:

- $BC_{p,j,k,y}$ is fixed at 1.54 Tonnes and is derived as an ex-ante calculation. It is calculated as $BC_{b,i,y}$ * ($\eta_{old,avg}/\eta_{new,avg,y}$). This parameter is a monitored parameter.

- $\eta_{j,k,y}$ is fixed as 0.90 and is a monitored parameter.

Determine Project Emissions (PEy):

Hence, Project Emissions over the project lifetime is determined as follows:

Table 13: Intermediate ER Calculations (Equation 7)

Equation 7	$PE_{energy,y} = BC_{p,j,k,y} \times N_{j,k,y} \times NCV_{p,j} \times n_{j,k,y} \times (EF_{b,i,CO2} \times f_{NRB,y} + EF_{b,i,nonCO2})$							
	$BC_{p,j,k,y}$	$N_{j,k,y}$	$NCV_{p,j}$	$n_{j,k,y}$	$EF_{b,i,CO2}$	$f_{NRB,y}$	$EF_{b,i,nonCO2}$	$PE_{energy,y}$
Year 1	1.54	100,000	0.0156	0.90	112	0.60	9.46	**165,349**
Year 2	1.54	95,000	0.0156	0.90	112	0.60	9.46	**157,081**
Year 3	1.54	90,250	0.0156	0.90	112	0.60	9.46	**149,227**
Year 4	1.54	85,738	0.0156	0.90	112	0.60	9.46	**141,766**
Year 5	1.54	81,451	0.0156	0.90	112	0.60	9.46	**134,678**
Year 6	1.54	77,378	0.0156	0.90	112	0.60	9.46	**127,944**
Year 7	1.54	73,509	0.0156	0.90	112	0.60	9.46	**121,547**
Year 8	1.54	69,834	0.0156	0.90	112	0.60	9.46	**115,469**
Year 9	1.54	66,342	0.0156	0.90	112	0.60	9.46	**109,696**
Year 10	1.54	63,025	0.0156	0.90	112	0.60	9.46	**104,211**

4.3 Leakage Emissions

As per VM0050 v1.0, Leakage emissions (LEy) is accounted as follows:

Leakage emissions associated with fossil fuel use:

a) Increased emissions from fossil fuel use by non-project participants

   An adjustment factor of 0.95 is applied to the GHG emission reductions. (i.e., 0.95 × (BEy − PEy))

As the project activity does not associate with renewable biomass ($LE_{RB,y}$), $LE_{RB,y}$ is '0'.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Net GHG emission reductions are calculated as follows:

$$ER_y = \left(BE_y - PE_y\right) \times 0.95 - LE_{RB,y} \quad (11)$$

Where:

ER_y = Emission reductions during year *y* (t CO_2e)

$LE_{RB,y}$ = Leakage emissions associated with use of renewable biomass during year *y* (t CO_2e)

Therefore, emissions reductions for the crediting period of the project activity are calculated as follows:

Table 14: Intermediate ER Calculations (Equation 11)

Equation 11	$ER_y = (BE_y - PE_y) \times 0.95 - LE_{RB,y}$				
	BE_y	PE_y	0.95	$LE_{RB,y}$	ER_y
Year 1	563,108	165,349	0.95	0	377,872
Year 2	534,953	157,081	0.95	0	358,978
Year 3	508,205	149,227	0.95	0	341,029
Year 4	482,795	141,766	0.95	0	323,978
Year 5	458,655	134,678	0.95	0	307,779
Year 6	435,723	127,944	0.95	0	292,390
Year 7	413,936	121,547	0.95	0	277,770
Year 8	393,240	115,469	0.95	0	263,882
Year 9	373,578	109,696	0.95	0	250,688
Year 10	354,899	104,211	0.95	0	238,153
Total	4,519,092	1,326,968	0.95	0	3,032,518

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
2024	154,276	45,301	5,449	103,526	0	103,526
2025	555,394	163,084	19,616	372,694	0	372,694
2026	527,625	154,929	18,635	354,061	0	354,061
2027	501,243	147,183	17,703	336,357	0	336,357
2028	476,181	139,824	16,818	319,539	0	319,539
2029	452,372	132,833	15,977	303,562	0	303,562
2030	429,754	126,191	15,178	288,385	0	288,385
2031	408,266	119,882	14,419	273,965	0	273,965
2032	387,853	113,887	13,698	260,268	0	260,268
2033	368,460	108,193	13,013	247,254	0	247,254
2034	257,666	75,660	9,100	172,906	0	172,906
Total	4,519,090	1,326,967	159,606	3,032,517	0	3,032,517

Note: A difference of +/- 1 in ERs is due to the rounding from project years to vintage years

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{b,i,CO2}$ $EF_{p,j,CO2}$
Data unit	tCO_2/TJ
Description	CO2 emission factor for fuel used by baseline device type i in the baseline scenario. CO2 emission factor for fuel used by project device type j in the project Scenario.
Source of data	2019 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 2, Stationary Combustion
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	Methodology Default
Purpose of data	Calculation of baseline and project emissions
Comments	N/A

Data / Parameter	$EF_{b,i,nonCO2}$ $EF_{p,j,nonCO2}$
Data unit	tCO2e/TJ
Description	Non-CO2 emission factor for fuel used by baseline device type i in the baseline scenario. Non-CO2 emission factor for fuel used by project device type j in the project scenario.
Source of data	2019 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 2, Stationary Combustion
Value applied	9.46

Justification of choice of data or description of measurement methods and procedures applied	Methodology Default
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	$NCV_{b,i}$ $NCV_{p,j}$
Data unit	TJ/tonne
Description	Net calorific value of baseline fuel used by baseline device type i Net calorific value of project fuel used by project device type j
Source of data	2019 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 1, Introduction
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	2019 IPCC default for wood fuel, 0.0156 TJ/tonne, based on the gross weight of the wood that is 'air-dried' may be used if fuel used in project device is woody biomass/renewable biomass.
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	$BC_{ex\text{-}ante,b,i}$
Data unit	Tonnes
Description	Ex-ante annual average quantity of fuel used per baseline device type i
Source of data	TEM 2025_Cookstoves Highlands region report_v3.pdf and audited by SPFEC Letter to TEM-Review of Cookstoves Highlands region report_17-01-2025.pdf
Value applied	5.24
Justification of choice of data or description of measurement methods and procedures applied	The quantity of fuel that would be used in the baseline scenario was determined through Option 1: Measurement Campaign conducted by a third-party consultant team contracted by TEM along with the baseline study survey concluded in November 2022. A total of 220 houses were selected for the baseline study

	with 110 houses each from the Southern Highlands Province and Western Highlands Province. The selected houses were provided with an individual code from 001 to 220. Of the total number of houses, 160 were used for control cooking tests (CCT) comparing firewood consumption using improved cookstoves (ICS, n=80) with traditional open fire cooking (OFC, n=80). At the end of the campaign, sampling challenges enabled the collection of firewood consumption data on 131 households in both provinces, of which 60 houses used ICS and the remaining 71 houses used traditional OFC. The results of the measurement campaign yielded a value of 5.97 tons as the firewood usage per year on a baseline stove (OFC) (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf). As the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 71 samples and achieved a precision value 12.36%, the lower bound value of **5.24 tons/year** is used after applying a conservativeness deduction as per Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0.
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	$\eta_{old,avg}$
Data unit	Fraction
Description	Weighted average efficiency of baseline devices that are replaced by project devices
Source of data	Methodology Default As per the applied methodology, for three-stone fire using firewood or a cookstove with no improved combustion air supply or flue gas ventilation, default value of 15%.
Value applied	0.15
Justification of choice of data or description of measurement methods and procedures applied	Methodology Default, CDM TOOL33_v2 During the baseline study TEM found that the prevailing pre-project device was a traditional three stone fire with no improved combustion air supply or flue gas ventilation, and thus, no efficiency tests were carried out (see file titled; TEM 2025_Cookstoves Highlands region report_v3.pdf).
Purpose of data	Calculation of baseline emissions

Comments	N/A

Data / Parameter	Hh_i $Hh_{j,k}$
Data unit	Equivalent standard male adults
Description	Average household size of the target population using device type i Average household size of the target population using device type j from batch k
Source of data	TEM 2025_Cookstoves Highlands region report_v3.pdf and audited by SPFEC Letter to TEM-Review of Cookstoves Highlands region report_17-01-2025.pdf
Value applied	4.85
Justification of choice of data or description of measurement methods and procedures applied	The baseline report conducted is in line with the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities v9.0 and has achieved confidence and precision of at least 90/10 for the target parameter of average household size.
Purpose of data	Estimation of average energy consumption when applying Option 1: Measurement campaign Cross-checking energy and fuel consumption values
Comments	N/A

Data / Parameter	$fNRB_{b,y}$
Data unit	Fraction
Description	Fraction of woody biomass that can be established as non-renewable biomass.
Source of data	TEM (2023). f_{NRB} calculation_Final (see letter titled: GeoNarem Letter for fNRB_Jan 23.pdf) audited by the Head of Forestry of PNG University of Technology (file titled: SPFEC Letter to TEM-Review of FnRB Calculations.pdf).
Value applied	0.60
Justification of choice of data or description of	Calculated as per the CDM Methodological Tool 30 Version 4: Calculation of the fraction of non-renewable biomass.

measurement methods and procedures applied	fNRB = NRB/(NRB+RB) = 81.24% Applying a discount of 26% = 60.12%

Parameter	Value	Source
NRB = H - RB	25,844,547	Calculated value
RB = MAI x (F – P)	5,967,605	Calculated value
MAI (Mean Annual Increment, t/ha/yr)	0.69	PNG's Second Biennial Update Report to UNFCC 2022, Table 2-8, pg 25
F (Extent of forest, ha)	35,963,272	Papua New Guinea Forest Authority (2019). Forest and Land Use Change in PNG 2000-2015, Table 7-1, p71
P (Extent of non-accessible area, ha)	27,296,091	Papua New Guinea Forest Authority (2019). Forest and Land use change in PNG 2000-2015, p35.
H = HW x N + CE + NE	31,812,152	Calculated value
HW (Average consumption of wood fuel per household, t)	6.8	Lit. reviews from Page et. al (2016), ACIAR (2013), Murphy et. al (2009), and TEM survey in project area (2023, unpublished)
N (Number of households in surveyed area)	197,869	
CE + NE (Commercial woody biomass consumption for energy	30,466,643	Data gathered from Papua New Guinea Forest Authority (2019). Forest and Land use change in PNG, 2000 -2015 and PNG's Second Biennial

	and non-energy applications respectively, t) | | Update Report to UNFCC 2022
Purpose of data	Calculation of emission reductions.
Comments	N/A

Data / Parameter	Life Span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project cookstove efficiency
Source of data	Manufacturer's specification
Value applied	10
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of data	Calculation of emission reductions.
Comments	N/A

5.2 Data and Parameters Monitored

Data / Parameter	$N_{j,k,y}$
Data unit	Number
Description	Number of commissioned project devices of type j from batch k in year y
Source of data	Distribution Ledger
Description of measurement methods and procedures to be applied	Distribution Ledger will be validated each time a new batch of ICS is distributed as part of the project instance. Example: If the project instance takes 3 years to fully distribute all ICSs, monitoring for this parameter will only be conducted for the first 3 years.

Frequency of monitoring/recording	Yearly, when there are new commissioned ICSs
Value applied	For ex-ante emission reduction calculation, it is assumed that the project will distribute up to 100,000 ICSs
Monitoring equipment	N/A
QA/QC procedures to be applied	The project implementor shall maintain a distribution record to calculate this parameter.
Purpose of data	Calculation of baseline and project emissions
Calculation method	The proportion of operational cookstoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value.
Comments	N/A

Data / Parameter	Date of commissioning of batch j
Data unit	Date
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in "batches" and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch
Source of data	Internal Log of Project device registrations.
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch
Value applied	30 September 2024
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of emission reductions.
Calculation method	N/A

Comments	To be reported in the monitoring report

Data / Parameter	Date of commissioning of project device i
Data unit	Date
Description	Actual date of commissioning of the project device
Source of data	Internal Log of Project device registrations.
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution
Value applied	Distribution is estimated to start on 23 September 2024.
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of emission reductions.
Calculation method	N/A
Comments	To be reported in the monitoring report

Data / Parameter	$n_{j,k,y}$
Data unit	Fraction
Description	Proportion of commissioned project devices of type j from batch k that are still being used regularly in year y
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	**Option 2: Surveys:** The total number of devices in operation compared to the total number distributed (according to the monitoring database) will be surveyed. Measured directly or based on a representative sample. Sampling standards shall be used for determining the sample size to achieve 95/10 confidence precision according to the latest

	version of Standard for sampling and surveys for CDM project activities and programme of activities. Or when efficient technology is available. <u>**Option 1: Stove Use Monitors (SUMs):**</u> Measured directly using stove use monitors (SUMs) in a sample of users according to the most recent version of the CDM Standard for Sampling and Surveys for Project Activities and Programmes of Activities and achieving 90/10 confidence precision for the proportion of devices in operation. The SUMs must confirm that the cookstove is frequently used and functional.
Frequency of monitoring/recording	Annually
Value applied	0.9 Assuming Option 2: Surveys is used, the usage rate will be capped at 90%. This is as per Clarification 2 of VM0050 v1.0 Correction and Clarifications[28].
Monitoring equipment	Monitoring Survey
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of baseline and project emissions
Calculation method	The proportion of operational cookstoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value.
Comments	N/A

Data / Parameter	$BC_{b,i,y}$
Data unit	Tonnes
Description	Fuel used per baseline device type i during year y
Source of data	**Option 1:** A measurement campaign following the Kitchen Performance Test Protocol in control households that do not participate in the project, established prior to validation as statistically equivalent to the pre-project conditions of project households regarding baseline fuel consumption. The control households are currently identified in the Koalilombo ward[29] of

[28] https://verra.org/wp-content/uploads/2025/02/CC_VM0050_v1.0_Feb2025.pdf

[29] GPS Coordinates: 6º 25’ 56.7” S 143º 54’ 11.2” E, https://maps.app.goo.gl/6J9omZzdJf47jdHL8

	the Southern Highlands Province. Based on PNG Census 2011[30], this ward has an estimated 341 houses and a population of approximately 1,992 people.
Description of measurement methods and procedures to be applied	**Option 1:** Follow-up baseline surveys must be conducted every five years in control households located in the Koalilombo ward of the Southern Highlands Province that do not participate in the project, established prior to validation as statistically equivalent to the pre-project conditions of project households regarding baseline fuel consumption. The measurement campaign must be updated where follow-up baseline surveys show that the fuels, fuel sources, or technologies used by the control group are no longer statistically equivalent to the pre-project conditions of project households. At the time of the follow-up survey, if the cooking practices of any number of these households are observed to be different than the pre-project conditions, they will be substituted with new control households for further monitoring. The measurement campaign must be designed, carried out, and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities version 9.0. The result must be scaled appropriately using the average household size to obtain the value of $BC_{b,i,y}$.
Frequency of monitoring/recording	Every five years
Value applied	For ex-ante calculations, 5.24 Tonnes/year was used.
Monitoring equipment	Monitoring Surveys
QA/QC procedures to be applied	The campaign must achieve confidence and precision of at least 90/10 for the target parameter of average daily fuel consumption per adult equivalent. Where the project does not achieve the target precision in a monitoring period, the project proponent must apply an appropriate conservative deduction as per Section 4 of the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities version 9.0.
Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	N/A

[30] National Statistics Office, Papua New Guinea National Population & Housing Census 2011, 'Count me in' (2011)

Data / Parameter	$\eta_{new,avg,y}$
Data unit	Fraction
Description	Weighted average efficiency of project devices in year y
Source of data	Monitoring Surveys, Manufacturer specifications on efficiency based on water boiling test (WBT).
Description of measurement methods and procedures to be applied	As per VM0050 v 1.0, the efficiency of project devices will be established via the following method: 1. Manufacturer-certified value that is determined via Water Boiling Test; The decrease in thermal efficiency of project devices from their respective batches due to aging must be accounted for during the monitoring period. As per VM0050, the loss in efficiency will be monitored annually through Standard Water Boiling Test campaigns from a respective sample of each batch. The actual loss rate is then measured and used.
Frequency of monitoring/recording	1) Recorded at the time of commissioning/distribution then; 2) Annually
Value applied	For ex-ante emission reduction calculation, a conservative value of 2% is estimated annually, although no losses were observed during surveys. Year 1: 0.52 Year 2: 0.51 Year 3: 0.50 Year 4: 0.49 Year 5: 0.48 Year 6: 0.47 Year 7: 0.46 Year 8: 0.45 Year 9: 0.44 Year 10: 0.43
Monitoring equipment	Monitoring survey, WBTs
QA/QC procedures to be applied	The weighted average efficiency of project devices is first established via a Manufacture-certified value that is determined via a Water-Boiling test. Subsequently, thermal efficiency due to aging is accounted annually via Standard Water Boiling Test campaigns as per AMS-II.G. v13.1, where the loss in efficiency

	must be monitored annually through the Standard Water Boiling Test campaign from a respective sample of the first batch. The actual loss rate is then measured and used. The following simplified approach may be used, when the efficient cookstoves are produced by a manufacturer with a recognized management system in place (e.g. ISO certification) to ensure that the individual equipment produced do not vary beyond the range of acceptance limits (e.g. characteristics such as materials, critical dimensions): 1. Conduct a sample test on three cookstoves with three tests conducted for each stove. The test can be carried out by project proponents by themselves or stove manufacturers; 2. If the standard deviation of the nine test results indicated above is very small and 90/10 precision requirement is met (in this case, the value of the t-distribution for 90 per cent confidence shall be used instead of Z value), the efficiency determined is acceptable, otherwise more sample tests would be required until 90/10 precision is met.
Purpose of data	Calculation of project emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	$BC_{p,j,k,y}$
Data unit	Tonnes
Description	Average quantity of fuel used by project device type j from batch k during year y
Source of data	Monitoring Surveys, KPTs
Description of measurement methods and procedures to be applied	$BC_{p,j,k,y}$ is estimated using Option 1: Kitchen Performance Test wherein a measurement campaign was conducted by a third-party consultant team contracted by TEM along with the baseline study survey concluded in November 2022. A total of 220 houses were selected for the baseline study with 110 houses each from the Southern Highlands Province and Western Highlands province. The selected houses were provided with an individual code from 001 to 220. Of the total number of houses, 160 were used for control cooking tests

	(CCT) comparing firewood consumption using improved cookstoves (ICS, n=80) with traditional open fire cooking (OFC, n=80). At the end of the campaign, sampling challenges enabled the collection of firewood consumption data on 131 households in both provinces, of which 60 houses used ICS and the remaining 71 houses used traditional OFC. The results of the measurement campaign yielded a value of 1.36 tons as the firewood usage per year on the project stove (ICS) (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf). As the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 60 samples and achieved a precision value 12.98%, the upper bound value of 1.54 tons/year is used after applying a conservativeness deduction as per Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0.
Frequency of monitoring/recording	Annually
Value applied	For ex-ante calculations, 1.54 Tonnes was used.
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	N/A

5.3 Monitoring Plan

Sampling Plan

The sampling plan is intended to enable the discovery of unbiased and reliable estimates of monitored parameter values used in the GHG emission reduction calculations. The sampling plan design is based on the CDM Guideline: Sampling and surveys for CDM project activities and programmes of activities, version 9.0, and will be designed to monitor the parameters listed in the Section above, which are required for calculation of the actual GHG emission reduction achieved by the project activity using ex post sampling survey.

The share of operating stoves and the continued use of pre-project devices will be determined based on sampling procedures as outlined below. The PP will be responsible for conducting the sampling surveys and maintaining a database with all operating stoves.

As per the section 6 of the Guideline for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 9.0, the sampling plan is the following:

1) Sampling Design

Due to the large number of ICS envisioned to be distributed as part of the project activity, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling will be undertaken as part of a project instance-wide Sampling Plan (by grouping and sampling across project activities) that is designed in line with the requirements of the "Sampling and surveys for CDM project activities and programme of activities", version 9.0.

1.1) Objective and Reliability Requirements

Sampling activities are defined at the Grouped Project-level and the sampling plan presented here will apply to assess parameter values in Project Instances to be included in the Grouped Project, either individually or in groups.

Table 15: Monitored Parameters

Parameter and Type	Parameter and Type	Indicative Timing	Frequency of Monitoring Required by Methodology	Methods to be Applied
$n_{j,k,y}$ (Proportion of commissioned project devices of type j from batch k that are still being used regularly in year y)	Fraction	Monitoring will likely occur at least once every 12 months	Annually	Field staff visual inspection of the ICS end users, and questionnaires
$BC_{b,i,y}$	Tonnes	Every five years	Every five years	Field staff visual inspection of the ICS end users, and questionnaires
$BC_{p,j,k,y}$	Tonnes	Biennially	Biennially	Kitchen Performance Test (KPT)
$\eta_{new,avg,y}$	Fraction	1. Recorded at the time of commissioning/distribution then; 2. Annually	Annually	1. Efficiency of project devices is established via a manufacturer-certified value.

				2. Decrease in thermal efficiency due to aging is monitored annually through the Standard Water Boiling Test campaign.

2.1) Target Population

- The target population for the proportion of ICSs still in operation ($n_{j,k,y}$) of this project are all households in the project database which are using fuel wood in ICS distributed under the project for cooking.

- The target population for fuel used per baseline device ($BC_{b,i,y}$) is set in control households that do not participate in the project, established prior to validation as statistically equivalent to the pre project conditions of project households regarding baseline fuel consumption.

- The target population for average quantity of fuel consumed by the project device ($BC_{p,j,k,y}$) are all households in the project database which are using fuel wood in ICSs distributed under the project for cooking.

- The target population for the weighted average efficiency of project devices ($\eta_{new,avg,y}$) is established via a Manufacture-certified value determined via Water-Boiling Test. Subsequently, the decrease in thermal efficiency of project devices from their respective batches due to aging must be accounted for during the monitoring period. As per AMS-II.G., the loss in efficiency has to be monitored annually through the Standard Water Boiling Test campaign from a respective sample of the first batch. The actual loss rate is then measured and used.

2.2) Sampling Frame

The projects are to be implemented in rural/semi-urban areas; thus, it is expected that the geographical locations do not have influence on the parameter of interest. Therefore, all above mentioned parameters can be assumed to be highly homogeneous for each ICS model regardless of how the end user group and distribution/installation location is defined.

1) Sampling frame for proportion of ICS in operation ($n_{j,k,y}$)

The sample frame refers to all the information sources on the Database. There are two primary mechanisms for data collection: the Registration Process for newly distributed/installed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) that will be used throughout the lifetime of the ICS. The detailed information collected from Registration Process is used to populate the stoves Database and the Monitoring Survey follows "Sampling and Surveys for CDM Project Activities and Programme of Activities", version 9.0.

As explained below (on section "Sampling Method"), to take the different characteristics of different project instances Implemented and ICS models into consideration, Project instances shall be grouped together to create a Primary Sampling Unit, which is homogenous. As per EB 86 Annex 04, Appendix-2, paragraph 1, for the use of a single sampling plan covering a group of projects, provided the homogeneity of population can be demonstrated, or differences are considered in the sample size calculation, a 95/5 confidence/precision is applied for biennial sampling.

The first step is to identify the Primary Sampling Units. Primary sampling units are project instances, which have:

Identification of Primary Sampling Units:

Primary sampling units are project instances, which have:

- The same ICS models

This project with the same ICS model can therefore be grouped together and form a Primary Sampling Unit.

2.3) Sampling Method

For each survey, the sampling method for monitored parameters is determined by using a Simple Random Sample of the total population of distributed ICS from records generated in the monitoring database and based on the precision requirements listed above, and as required by the methodology and guidance in the Sampling Standard.

For parameter $\eta_{new,avg,y}$, loss in efficiency of the project devices due to aging during the monitoring period will be accounted for using one the following methods set out in VM0050 v1.0:

1. Standard Water Boiling Test campaigns as per AMS-II.G. v13.1, where the loss in efficiency must be monitored annually through the Standard Water Boiling Test campaign from a respective sample of the first batch. The actual loss rate is then measured and used; or

2. A linear decrease approach, applying a default schedule of linearly decreasing efficiency up to the terminal efficiency (assumed to be 25%) through the lifespan of the project device.

In the event that Option 1 is not available, Option 2 will be used.

To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique serial number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the Project Participant.

To determine the parameters, sampling will involve the following approaches:

$N_{j,k,y}$: Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use.

$n_{j,k,y}$: The total number of devices in operation compared to the total number distributed (according to the monitoring database) will be surveyed. Measured directly or based on a representative sample. The Sampling standard shall be used for determining the sample size to achieve 95/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and programme of activities version 9.0.

$BC_{b,i,y}$: A measurement campaign following the Kitchen Performance Test Protocol in control households that do not participate in the project, established prior to validation as statistically equivalent to the pre-project conditions of project households regarding baseline fuel consumption.

$BC_{p,j,k,y}$: A Kitchen Performance Test protocol will be designed, carried out, and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities version 9.0. The campaign must achieve confidence and precision of at least 90/10 for the target parameter of average daily fuel consumption per adult equivalent. The result must be scaled appropriately using the average household size to obtain the value of $BC_{p,j,k,y}$.

$\eta_{new,avg,y}$: The weighted average efficiency of project devices is first established via a Manufacture-certified value that is determined via a Water-Boiling test. Subsequently, thermal efficiency due to aging is accounted annually via Standard Water Boiling Test campaigns as per AMS-II.G. v13.1, where the loss in efficiency must be monitored annually through the Standard Water Boiling Test campaign from a respective sample of the first batch. The actual loss rate is then measured and used.

The following simplified approach may be used when the efficient cookstoves are produced by a manufacturer with a recognized management system in place (e.g. ISO certification) to ensure that the individual equipment produced do not vary beyond the range of acceptance limits (e.g. characteristics such as materials, critical dimensions):

- Conduct a sample test on three cookstoves with three tests conducted for each stove. The test can be carried out by project proponents by themselves or stove manufacturers.
- If the standard deviation of the nine test results indicated above is very small and 90/10 precision requirement is met (in this case, the value of the t-distribution for 90 per cent confidence shall be used instead of Z value), the efficiency determined is acceptable, otherwise more sample tests would be required until 90/10 precision is met.

Using the formulas in the section “Sample Size” below, the Project Participant will randomly sample the required number of ICSs from the primary sampling units.

2.4) Sample Size

The sample size for each monitoring activity will be the total number of ICS distributed and included in the monitoring database. The minimum sample size for each sample frame must achieve the 90/10 confidence/precision. The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error.

In line with sampling guidelines, all can be sampled in a single survey with a random sample of households using the above-described confidence/precision levels depending on annual or biennial monitoring frequency.

As per Guidelines for Sampling and surveys for CDM project activities and programmes of activities version 09.0 paragraph 13, there are different ways available to obtain the estimates of the parameter of interest:

1. The type of parameter of interest, that is, mean value or proportion value.
2. The target value, that is, the expected value of the parameter, which should be determined using the project participants’ or the coordinating/managing entity’s knowledge and experience.
3. Expected variance (or standard deviation) for that measure in the sample, based on the results from similar studies including other similar CDM project activities or previous monitoring periods, pilot studies, or from the project planner’s own knowledge of the data.

For the registration purpose of project, option 2 shall be applied. For the first monitoring period, values from a pilot shall be applied. For the following monitoring periods, the estimates shall be adjusted considering the results of the previous monitoring period(s) or the result from a recent pilot study, which is conducted after the previous monitoring periods.

To estimate the sample size for parameters $n_{y,i,j}$ and μ_y the following equation is used:

Table 16: Sample size calculation

$$n \geq \frac{1.96^2 \, N \times p\,(1-p)}{(N-1) \times 0.1^2 x p^2 + 1.96^2 x\, p(1-p)}$$

n	=	Sample size
N	=	Population size (Total number of households/ICS)
p	=	Expected proportion

In case the resulting sample size to achieve the desired confidence/precision levels is smaller than 30 ICSs, a minimum sample size of 30 shall be chosen when the parameter of interest is a proportion in line with para 14 of Standard: Sampling and surveys for CDM project activities and programmes of activities version 9.0.

TEM envisages distribution of 100,000 ICSs in Year 1. Hence, population size, N, is taken as 100,000 household/ICS (assuming 1 ICS per household).

With an expected ICS loss rate of 5%, it is expected that 95% of ICSs will be operational, hence the expected proportion p for $N_{j,k,y}$ is taken as 0.95.

Oversampling will be conducted to ensure that the required level of precision is met, not only to compensate for any attrition, outliers or non-response associated with the sample, but also to prevent a situation at the analysis stage where the required reliability is not achieved, and additional sampling efforts would be required.

Data collection of ICS end-user

Project proponent must gather the necessary information to identify households using its ICS during the project. To facilitate this process, each ICS will be assigned a USN. This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address / GPS of ICS household
- Phone number of ICS user or household, where available.
- Stove model
- Date of distribution/installation
- ICS serial number
- Retailer/distributer information

The information collected from the end-user will be transferred to an electronic database, which will be updated regularly. The distribution record carries all distribution information including the traditional stove type (charcoal cookstove/firewood cookstove) used prior to ICS installation, name of representative member of household, phone number (if available and user-permitted for sharing), address etc. Likewise, monitoring records are transferred from each monitoring organization, if any, to the PP. PP will ensure that appropriate records are maintained for the project activity.

A record keeping system and unique identification of the project is made possible by user's information collected through the End-User Agreement and compiled in distribution records. This will include end-user name, identification number and address/location of the user's house, cookstove unique serial code and distribution date. The record keeping system will ensure that each ICS can be traced to the project activity to avoid double counting.

The Monitoring Plan applied in this project involves several key elements that ensure that the PP have high-quality, unbiased, and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating emissions reductions on the basis of the amount of non-renewable biomass saved by the ICS in the project activity.

The key elements are the following:

- Project database management
- Sample Plan for the Monitoring Survey
- Data Quality, Consistency and Duplication Checks
- Monitoring Reporting

The TEM PNG's Country Manager will have overall responsibility for the implementation of the monitoring plan. The information collected will be stored in the electronic database and will be updated on ongoing basis by trained staff.

Data collection will be conducted by a dedicated monitoring team of trained individuals, from TEM PNG. All technical staff responsible for installation and maintenance of the stoves will be trained in terms of the understanding the requirements on the monitoring system.

2.5) Organizational Structure

The project will be managed by TEM as the project participant. The roles and responsibilities are shown in the table below:

Table 17: Organizational Structure

Person/Entity	Responsibility
TEM PNG Country Manager, Field Coordinator and Field Assistants	• Oversee operation of distribution centres; execution of set up activities; works with project manager on all planning. • Execution of set up activities such as recruitment and training, reporting of monitoring data. • Logistics Manager: planning; identification of target households; contractor management; overall day to day management of installation staff; weekly and monthly reporting. • Data administrators: monitoring database management; accounting; data reconciliations; monthly reporting • Pre- & Post-distribution data collection: conveying project messages; distributing the project; signing up householders wanting a cookstove; sign up data capture; distribution data capture • Distribution team: management of distribution process; ensuring quality cookstove distributions. • Monitoring team: gathering compliance monitoring data; gathering marketing data; data input
TEM Director of International Projects	• Project planning and management; issue and risk management. • Oversight of management system & signoff on ICS procurement and monitoring reports, review of competencies of team members • Technical review team: technical review of process and documentation and monitoring reports. • Compliance Manager/Consultant: writing PD & monitoring reports, ensuring compliance with VCS rules.

Elaboration of Tasks:

Table 18: Elaboration of tasks

Person/Entity	Task	Responsibility
TEM Country Manager, Field Coordinator and Field Assistants	Manufacturing and logistics	Depending on the cookstove model, complete stoves or components for the stoves are: • Manufactured (some imported, others produced locally) by a cookstove manufacturer. • Stoves are distributed to warehouses within project region. • Coordinate the distribution of stoves to recipient households
TEM Country Manager, Field Coordinator and Field Assistants	Household identification	• Work with local community leaders and other partners to help identify recipient households suitable for the distribution of a cookstove. • Establish partnership with community leaders, NGOs and other local organizations, local partner will initiate a communication process to ensure that households understand the benefits of the stoves, that cultural issues are addressed and that users are trained in the optimal use and performance of the cookstove. • TEM team prior distribution will visit recipient communities in each region and ensure recipients understand and acknowledge the conditions for participation in the project. • Ensure each cookstove is assigned a unique distribution number chronologically to avoid any double counting thought the use of an electronic database
TEM Country Manager, Field Coordinator and Field Assistants	Distribution	• TEM team train cookstove distribution teams to distribute stoves. • Coordinate the receipt of stoves and components in the distribution process.

		• TEM team will be trained in the distribution of the cookstove to a standardized delivery and communication procedure. • TEM team will be responsible for physically distributing the ICSs to the recipient household
TEM Country Manager, Field Coordinator and Field Assistants	Data Capture	• The information collected from the TEM team during the distribution of ICS, from end-users and interactions on the monitoring phase of the project will be transferred to an electronic database, which will be updated regularly. The database will include a unique serial number for each cookstove to avoid double counting. • TEM PNG team checks the quality of ICS and record keeping work. If the work is satisfactory, distribution data is collected by the TEM PNG team
TEM Director of International Projects	Monitoring	Monitoring activities will be conducted as follows: • Surveys completed in the field by trained local monitoring teams. • Data captured by the monitoring teams is passed to the TEM Head of Science and Sustainability • Data is checked for completeness, consistency, and accuracy. • Project manager summarizes data in a report to TEM Director of International Projects • Writing of monitoring reports for each monitoring period. • Technical review by in-house technical team

2.6) QA/QC

In order to minimize sampling errors, such as non-responses and errors, the field team will practice over-sampling from the population to ensure a total number of respondents that meets the required level of precision. This will ensure the integrity of the sample group and maintains the randomness of participant selection. All samples' groups will be re-selected for each monitoring period / year, as appropriate for the parameter in question.

If the required level of precision is not achieved, then further surveys may be completed, or the lower bound value adopted in the calculations.

Where a survey may not be completed, or where there is a non-response, the reasons shall be clearly documented in the survey questionnaire.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No commercially sensitive information has been excluded from the public version of the project description.

APPENDIX II: LSC OUTREACH

1. **Newspaper Advertisements:**

There were two newspapers published nationally in PNG. As the project activity would be a grouped project activity, TEM saw the importance of announcing the to the public and informing stakeholders of the project, as well as briefing them on the upcoming local stakeholder consultations happening in the Highlands Provinces. The following table shows the date of when these advertisements were published in the newspaper:

S/N	Publishing House	Advertisement Published on: (date)	Supporting Documentation
1	Post Courier Newspaper	14/02/2023	Invoices from respective newspapers for publishing, picture of advertisement in newspapers

2. **Letters of Notification to Provincial Governments:**

Letters of Notification were also hand delivered to provincial government administrative assistants to brief them on the project activity as well as the consequent co-benefits to their local community that came with the project. Together with the Letter of Notification, a project brief was also attached to aid their understanding. The following table shows the dates of when the letters of no-objection were received from the provincial governments:

S/N	Province	Document Received on: (date)	Supporting Documentation
1	Western Highlands Province	13/02/2023	Letter of No-Objection
2	Jiwaka Province	15/02/2023	
3	Eastern Highlands Province	16/02/2023	
4	Southern Highlands Province	08/03/2023	

During TEM's consultations with the provincial governments, the administrative assistants expressed a high level of interest in the project and were supportive of its implementation.

3. **Radio Broadcasts**

To further TEM's outreach into the Highlands Provinces regarding the LSCs, radio broadcasts were utilized. All radio broadcasts were done by the National Broadcasting Commission which then broadcasted TEM's announcements to individual provinces at a frequency of 1 time a day, over 10 days. Each province-level broadcast would then inform the specific province of the date, location and time of which the LSC was happening within their area. The following table shows the start and end dates of these broadcasts:

S/N	Province	Start of Broadcast: (date)	End of Broadcast: (date)
1	Western Highlands Province	29/02/2023	12/03/2023
2	Jiwaka Province	29/02/2023	12/03/2023

The radio broadcast could only be done for 2 highlands provinces as it was found that the radio stations for the remaining two provinces had been shut down.

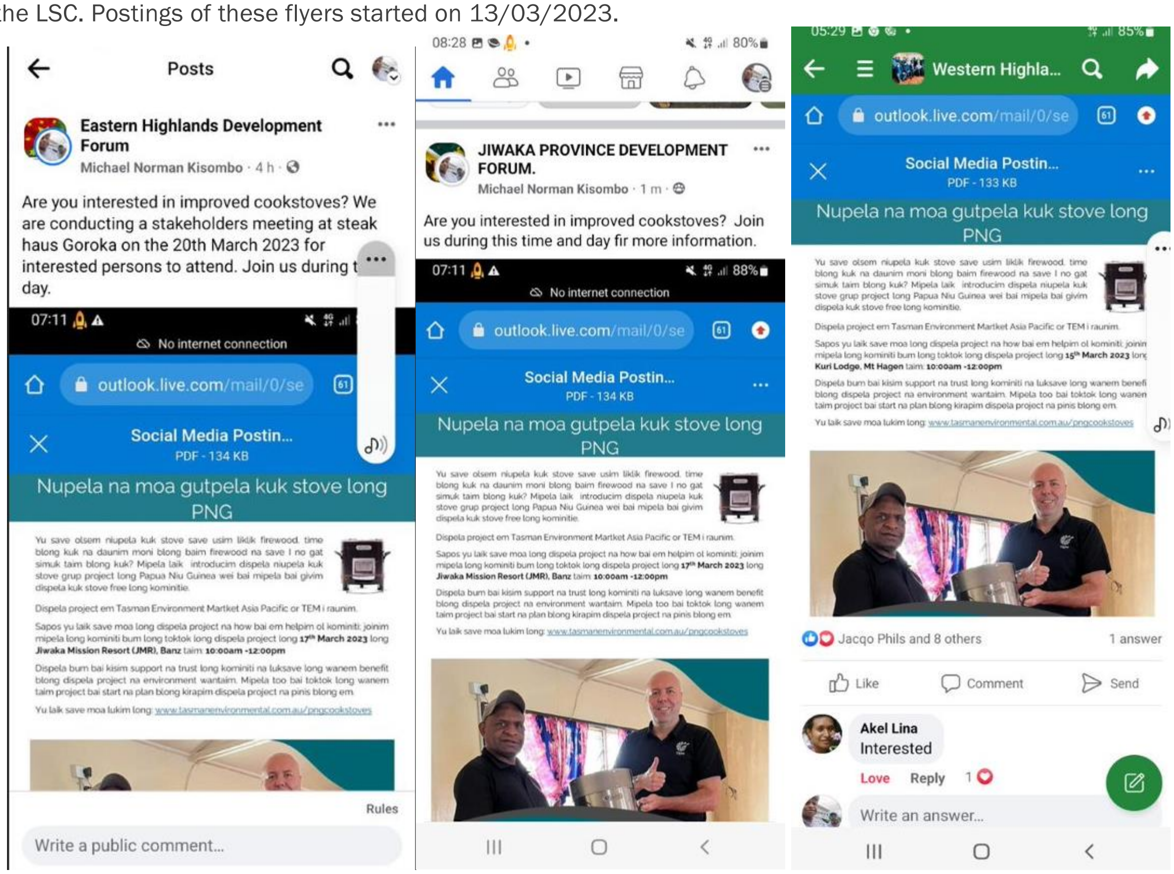
4. **Invitation to Churches**

TEM is aware that churches and church leaders play an important role in the local community within PNG. As such, invitations to churches were also hand delivered to churches within each province to invite church leaders and elders to participate in the LSC, as well as spread the word about the importance of the LSC within their community by word-of-mouth. The following table shows the dates these invitations were delivered to churches within the provinces:

S/N	Province	Delivery of Invitation: (date)	Supporting Documentation
1	Southern Highlands Province	10/02/2023	Acknowledgement of Receipt by Church Bodies: SHP
2	Western Highlands Province	15/03/2023	Acknowledgement of Receipt by Church Bodies: WHP, Jiwaka, EHP
3	Jiwaka Province	17/03/2023	
4	Eastern Highlands Province	20/03/2023	

5. Social Media Postings

TEM also understood that the majority of the stakeholders received news of events happening within their area through social media. For this, flyers depicting the details of the LSC, and introduction of the project were developed and posted on each province's dedicated Facebook development forums to further boost stakeholder awareness of the LSC. Postings of these flyers started on 13/03/2023.

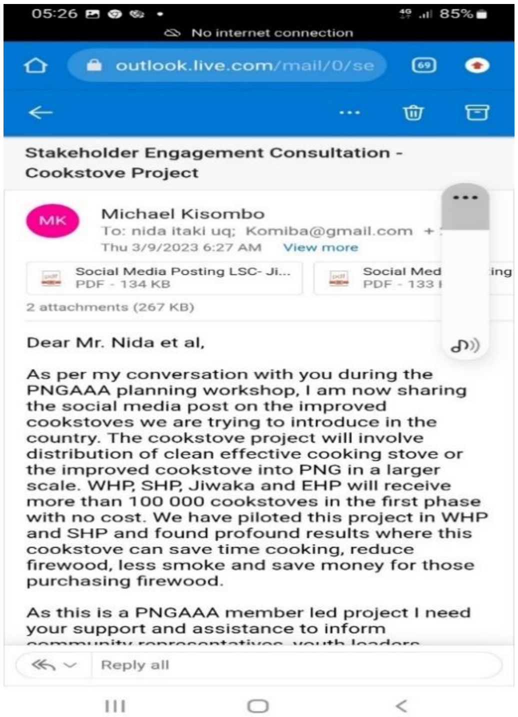


6. Email to PNG Australia Alumni Association (PNGAAA) members

From within TEM's local partner's connections, TEM also reached out to members from WHP and EHP via email. The table below shows the dates these emails were sent out:

S/N	Province	Date of Email Sent: (date)
1	WHP	09/03/2023
2	EHP	16/03/2023

Screenshots of the emails sent out can also be found below:


05:26
No internet connection
outlook.live.com/mail/0/se

Stakeholder Engagement Consultation - Cookstove Project

Michael Kisombo
To: nida itaki uq; Komiba@gmail.com +
Thu 3/9/2023 6:27 AM View more

Social Media Posting LSC- Ji... PDF - 134 KB
Social Med...ing PDF - 133 ...

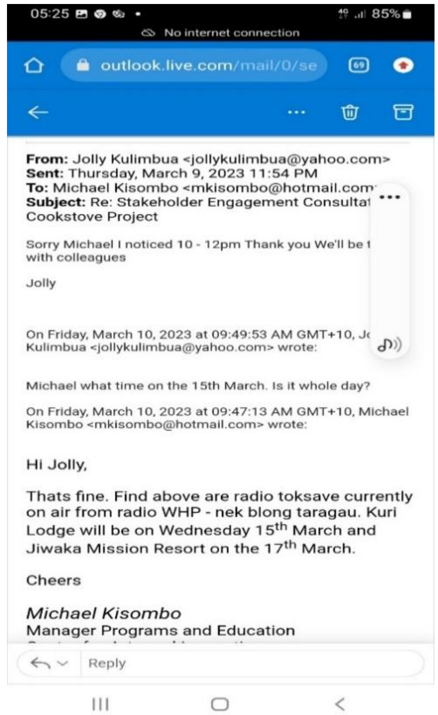
2 attachments (267 KB)

Dear Mr. Nida et al,

As per my conversation with you during the PNGAAA planning workshop, I am now sharing the social media post on the improved cookstoves we are trying to introduce in the country. The cookstove project will involve distribution of clean effective cooking stove or the improved cookstove into PNG in a larger scale. WHP, SHP, Jiwaka and EHP will receive more than 100 000 cookstoves in the first phase with no cost. We have piloted this project in WHP and SHP and found profound results where this cookstove can save time cooking, reduce firewood, less smoke and save money for those purchasing firewood.

As this is a PNGAAA member led project I need your support and assistance to inform

Reply all


05:25
No internet connection
outlook.live.com/mail/0/se

From: Jolly Kulimbua <jollykulimbua@yahoo.com>
Sent: Thursday, March 9, 2023 11:54 PM
To: Michael Kisombo <mkisombo@hotmail.com>
Subject: Re: Stakeholder Engagement Consultat... Cookstove Project

Sorry Michael I noticed 10 - 12pm Thank you We'll be t... with colleagues

Jolly

On Friday, March 10, 2023 at 09:49:53 AM GMT+10, J... Kulimbua <jollykulimbua@yahoo.com> wrote:

Michael what time on the 15th March. Is it whole day?

On Friday, March 10, 2023 at 09:47:13 AM GMT+10, Michael Kisombo <mkisombo@hotmail.com> wrote:

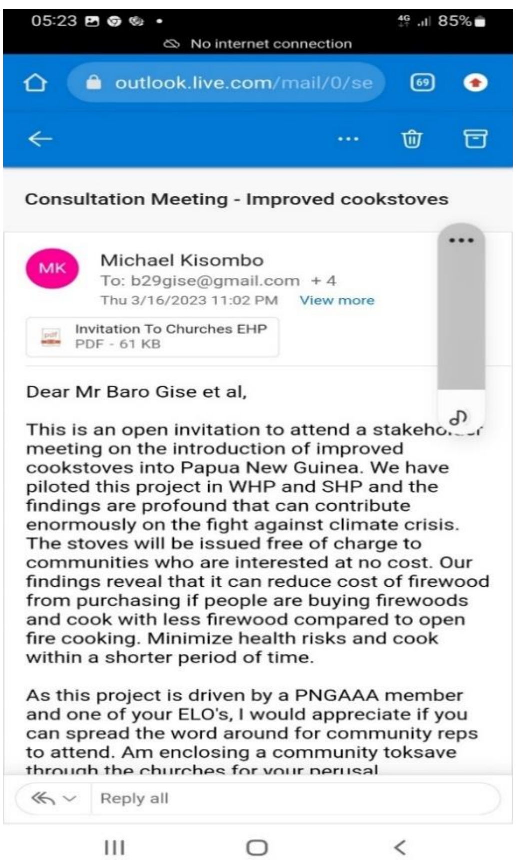
Hi Jolly,

Thats fine. Find above are radio toksave currently on air from radio WHP - nek blong taragau. Kuri Lodge will be on Wednesday 15th March and Jiwaka Mission Resort on the 17th March.

Cheers

Michael Kisombo
Manager Programs and Education

Reply


05:23
No internet connection
outlook.live.com/mail/0/se

Consultation Meeting - Improved cookstoves

Michael Kisombo
To: b29gise@gmail.com +4
Thu 3/16/2023 11:02 PM View more

Invitation To Churches EHP
PDF - 61 KB

Dear Mr Baro Gise et al,

This is an open invitation to attend a stakeho... meeting on the introduction of improved cookstoves into Papua New Guinea. We have piloted this project in WHP and SHP and the findings are profound that can contribute enormously on the fight against climate crisis. The stoves will be issued free of charge to communities who are interested at no cost. Our findings reveal that it can reduce cost of firewood from purchasing if people are buying firewoods and cook with less firewood compared to open fire cooking. Minimize health risks and cook within a shorter period of time.

As this project is driven by a PNGAAA member and one of your ELO's, I would appreciate if you can spread the word around for community reps to attend. Am enclosing a community toksave through the churches for your perusal

Reply all

APPENDIX III: STAKEHOLDER FEEDBACK

1. **Southern Highlands Province:**

Questions:

S/N	Question	Answer
1	Ms Julie Ponaria (a teacher from SHP) enquired that there are many big countries contributing to the destruction of the ozone layer and now trying to give (us) cookstoves to stop carbon emissions. She asked if there is a story behind this small cookstove?	Yes, there are many industrial countries that are destroying the ozone layer because they have big factories and industries. We (people of PNG) also do so when cutting down trees for farming and cooking etc… This project is to reduce carbon emissions by using less firewood for daily needs. Hence, saving trees that consume carbon and restore them. That is the story behind this project.
2	Ps Joseph Yapasi of Tente Ward 2 commented that 50 000 cookstoves is not sufficient to cater for all households in Southern Highlands and there may be risks when distribution. Therefore, he suggested that distribution be more targeted. E.g distribute through church networks or distribute to a particular district first and later target other districts.	It was noted for TEM to decide.
3	Ms Regina Kamerepa pointed out that since the cookstoves are going to be issued free, people are likely to rush to get one. Hence, can also steal or pull them from the distributors during the distribution. And asked how TEM would manage that scenario?	This is a threat we are aware of and are working on a strategy on how this can be mitigated during distribution. Currently, it is planned that security will be present during the distribution process. End-users will also be advised to keep their cookstoves in a safe place to deter theft of the device.

Reflections:

S/N	Reflections
1	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.
2	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.
3	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.
4	Mr. Peter Wari who is in charge of climate change in the province acknowledged the project as it was really going to benefit the community. He further stressed about the importance of the project activity and encouraged the stakeholders to support the project.
5	Most stakeholders mentioned that there is an increase in the usage of firewood and most of their trees (caussarina/yar) are diminishing because people are either chopping them to sell as there is a high demand or for household use. Therefore, the introduction of such cookstoves will be relieving for them.
6	Most stakeholders mentioned that there is an increase in the usage of firewood and most of their trees (caussarina/yar) are diminishing because people are either chopping them to sell as there is a high demand or for household use. Therefore, the introduction of such cookstoves will be relieving for them.
7	Another youth mentioned that he provides firewood for three separate houses as most are windows who have no husbands. With the new cookstove, he believes will relieve him from time and struggles he is facing foraging for firewood for all he’s mothers.

8	A Health worker emphasized that inhaling of smoke using normal open fire cooking is contributing to a lot of respiratory illnesses and with the introduction of the improved cookstove she believes will minimize such especially mothers and girls.
9	Besides the above reflections and questions, there were quiet a good number of people giving positive comments and supported the project activity. They even asked to increase the volume from 50 000 cookstoves to enough to cater for all households the province.

2. **Western Highlands Province:**

Questions:

S/N	Question	Answer
1	Mr. Frank Kuri Cr from Mul Baiyer asked, since it's going to affect the rural communities, can the councilors get involved?	Yes, Councilors will get involved to identify heads of households but we do not want this project to be politicized, either be councilors or local MPs.
2	Mr. Nida L Taki, a lecturer at the Agriculture College, Mt Hagen asked if there are further studies made to determine the level of smoke that is given out during normal open fire cooking to see how much carbon emissions are produced.	That's an interesting question. However, this project used observation method to observe the level of smoke when cooking both during the control cooking test and open fire cooking.
3	Mr Dickson Wia stressed that firewood is not only used for cooking in the highlands but for other purposes too. Will this cookstove provide for all?	Yes, we use firewood for many other purposes like for heat, light, storytelling etc. This cookstove is for cooking and it can also heat up the house.
4	Rex Parak enquired if the design of the cookstove can be made locally?	The brand and model of cookstove was chosen specifically for its high efficiency. TEM will consider a local brand in the future if it fits the specifications required to yield the highest efficiency in the combustion of firewood.

Reflections:

S/N	Reflections
1	The District Administrator for Dei District thanked the team for introducing this new innovative idea of cooking as people in his electorate are running out of firewood. He welcomed the project initiative and asked to include all rural communities in his electorate.
2	Those who came from the urban centers also requested that they should be given priority as they are using a lot of money to purchase firewood and its already becoming a burden.
3	One of the participants mentioned that the project is very beneficial as it saves time foraging for firewood and financial relief in buying firewood.
4	A youth mentioned that this project will benefit the youth as he sees that firewood is already running out.
5	Another supported by saying, the initiative taken by TEM is indeed very vital to reduce carbon monoxide as well as relieving economic burden and reduce health related issues. She suggested that it would be good to involve the councilors.

6	One said, this project must not stop after 10 years. It must continue till the world ends.
7	A health worker supported the project by saying, as a health worker this project if rolled out effectively will help reduce the number of sicknesses like pneumonia, chronic obstructive lung disease and eye diseases as well.

3. **Jiwaka Province:**

Questions:

S/N	Question	Answer
1	Mr. Lumun Mingal asked how can PNG be compensated well since we are less carbon emitters yet are asked to further reduce the emissions through such cookstove project?	This cookstove project aims to contribute to the wellbeing of the stakeholders in PNG by lessening the burden of purchasing or collecting firewood and improving the respiratory health of end-users exposed to hazardous smoke from the burning of firewood. Cookstoves being distributed for free is paid for through carbon finance and is one method of compensating stakeholders using the carbon credit scheme.

Reflections:

S/N	Reflections
1	A local landowner thanked TEM for this initiative and welcomed everyone to JMR. Having said that he suggested that this project be based there as it is the model village for Jiwaka and must be made the main distribution point.
2	Another said, there are many NGOs like IOM, UNICEF, etc. who come and deliver similar community-based projects but we the people seem to sell those items. He was worried the cookstoves might be sold in a similar manner and appealed to other participants to discourage such practices.
3	A female teacher mentioned that as a teacher she moves from one area to another with kerosine stoves and with the introduction of the improved cooking stove, hers and other teachers would benefit greatly. Not only that but emphasized that they usually ask their students to bring firewood for them once a week and this project will ease that burden.
4	Another person suggested that every household in Jiwaka must receive a cookstove as they are facing difficulties in looking for firewood.
5	Sammy Buka of Banz mentioned that the project is a bottom up where local communities will benefit. He said it's a sickness free project.
6	There were other supportive comments like, TEM must continue this project to other provinces in PNG, delivered through churches, schools and LLG wards.

7	When asked if there are risks involved like those of you selling firewood, would this project affect you? Everyone said no, not at all as they see there is a lot of benefit to the rural communities.
8.	Someone also wanted more than 50 000 cookstoves in Jiwaka as they believe they are good people.

4. **Eastern Highlands Province:**

Questions:

S/N	Question	Answer
1	Mr Rex Yagusa mentioned that there is a high possibility of the cookstoves being stolen by thieves or destroyed by natural and man-made disasters and asked in such a case, how can we register these complains or cases?	Currently, it is planned that security will be present during the distribution process. End-users will also be advised to keep their cookstoves in a safe place to deter theft of the device.

Reflections:

S/N	Reflections
1	One of the ladies (video graphed) mentioned that this cookstove will lessen the burden for those mothers in the rural setting. She said if they don't sell, there is no money to purchase firewood so they would use plastics and cardboard to cook. This project will really relieve those struggling mothers living in the urban areas.
2	Another suggested having the distribution point centralized and allowing only genuine households to collect. Later field officers can be sent to their locations for registration such as GPS points etc.
3	Someone in the group mentioned that economic benefits are huge as they will be working as the project team to distribute.
4	Those who attended expressed support for the project and demanded if more is distributed in EHP given their population density.
5	No risks or harm mentioned when implementing this project.
6	One participant mentioned that tree-cutting is happening almost every day. More supplies to meet population will help reduce carbon emission.